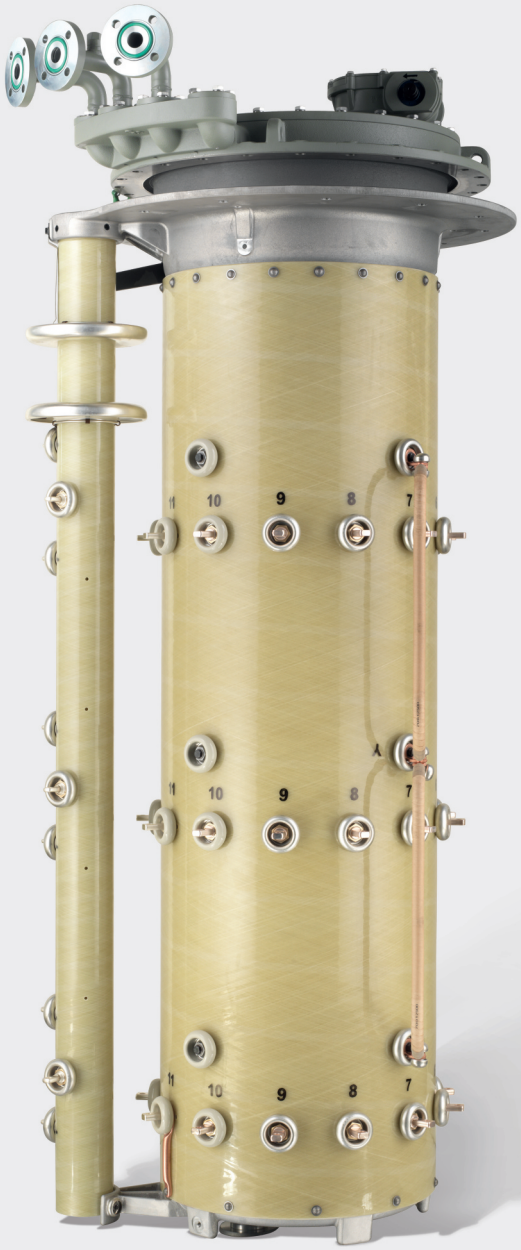
Technical data
VACUTAP® VV®. On-load tap-changer

1800203/06 EN



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Generally the information provided and agreements made when processing the individual quotations and orders are binding.
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The original operating instructions were written in German.

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1 General

1.1 Summary of the technical data

	VV III 250 Y	VV III 250 D	VV III 400 Y	VV III 400 D	VV III 600 Y	VV III 600 D	VV I 401[3]
Number of phases and application	3, neutral point	3, at any point in the winding	3, neutral point	3, at any point in the winding	3, neutral point	3, at any point in the winding	1, at any point in the winding
Max. rated through-current I_{um} [A]	250		400		600		400
Rated short-time current [kA]	6		6		6		6
Rated duration of short-circuits [s]	3		3		3		3
Rated peak withstand current [kA]	15		15		15		15
Rated step voltage U_i [V]	2 000		2 000...1 700[1]		2 000...1 000[1]		2 000...1 700[1]
Switching capacity [kVA]	see Pst/IU diagram [▶Section 2.1, Page 12]						
Rated frequency [Hz]	50...60						
Operating positions	without change-over selector: max. 12 with change-over selector: max. 23						with change-over selector: max. 23
Highest voltage for equipment U_m [kV][2]	40/76/123	40/76/145[4]	40/76/123	40/76/145[4]	40/76/123	40/76/145[4]	76/145[4]
Rated lightning impulse withstand voltage [kV/1.2/50]	200/350	200/350/650	200/350	200/350/650	200/350	200/350/650	350/650
Rated short-duration power frequency withstand voltage[kV/50 Hz/1 min]	70/140	70/140/275	70/140	70/140/275	70/140	70/140/275	140/275
Rated withstand voltages of the internal insulation	see Rated withstand voltage of inner insulation [▶Section 2.3, Page 13]						
Oil compartment	Pressure-tight up to 0.3 bar permanent differential pressure (test pressure 0.6 bar), head and cover of the on-load tap-changer are vacuum-proof						
Temperature range	The VACUTAP® VV on-load tap-changer can be operated in the rated load range at oil temperatures of between -25 °C and+105 °C						
Tap-change supervisory control	optional, see drawing 733452						
Pressure relief device MPREC	optional, see drawing 737780						

[1] see also U_i/I_u diagram
[2] In accordance with VDE0111 Part 1: Effective value of the conductor-conductor voltage for which equipment is dimensioned in terms of its insulation
[3] 600 A model on request
[4] A max. operating voltage of 132 kV + 15 % = 151.8 kV is admissible if the test voltages for the 145 kV series are not exceeded

Table 1: Summary of the technical data

	VV III 40 kV		VV III 76/123 kV		VV III 145 kV		VV I 76 kV		VV I 145 kV	
Weight without oil [kg]	approx. 240		approx. 255		approx. 280		approx. 200		approx. 215	
Displacement volume [dm^3]	295		328		405		148		178	
Oil filling quantity V_s and minimum oil conservator volume ΔV[1] [dm^3]	V_s	ΔV	V_s	ΔV	V_s	ΔV	V_s	ΔV	V_s	ΔV
	240	30	265	33	325	39	125	16	145	19
Oil suction pipe	optional				optional[2]		optional		optional[2]	

[1] Applies for oil temperature ϑ = -30 °C...+100 °C

[2] Applies for 145 kV series: Oil suction pipe not to be used together with oil filter unit

Table 2: Summary of the technical data

1.2 Required minimum height of the oil conservator

The height of the oil conservator is calculated from the upper edge of the on-load tap-changer cover to the oil level in the oil conservator (see following image).

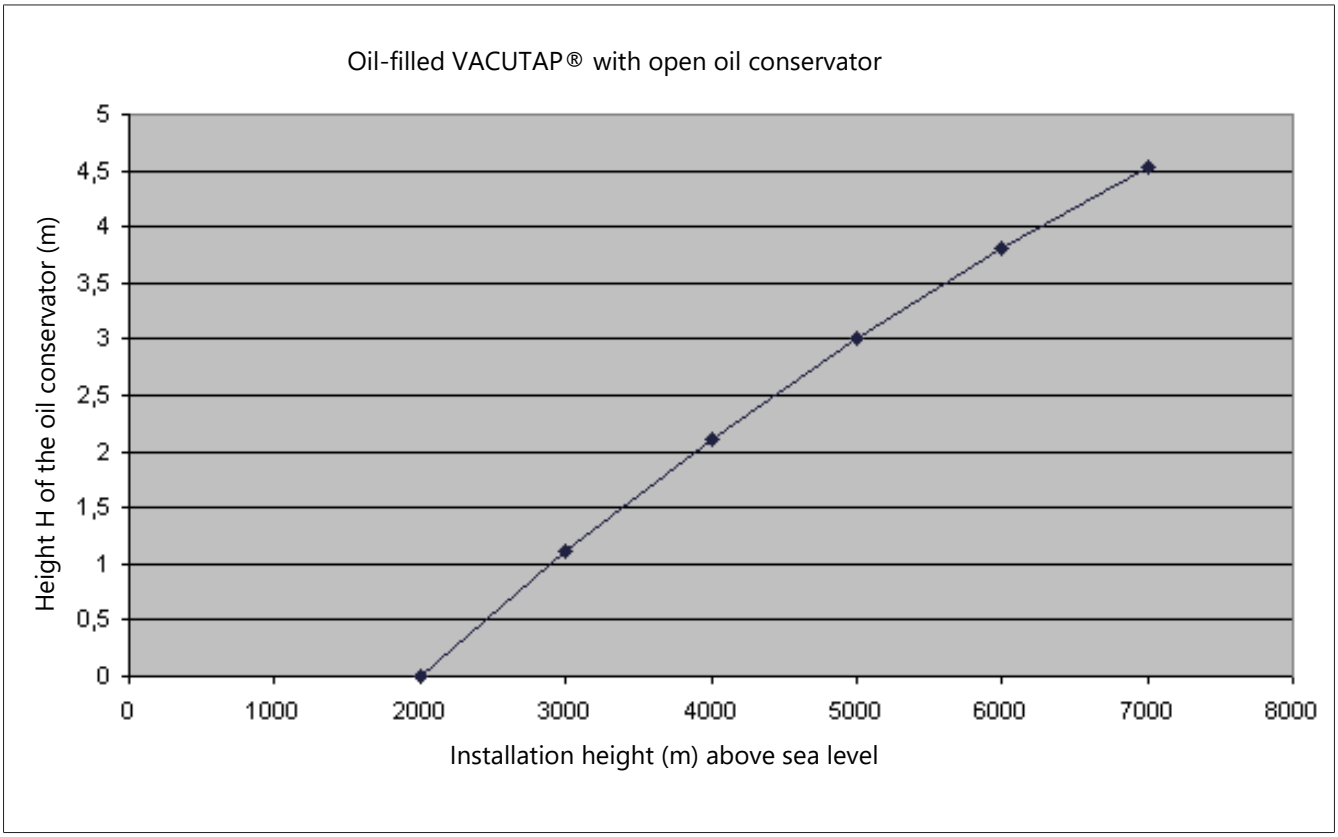


Figure 1: Required minimum height of the oil conservator

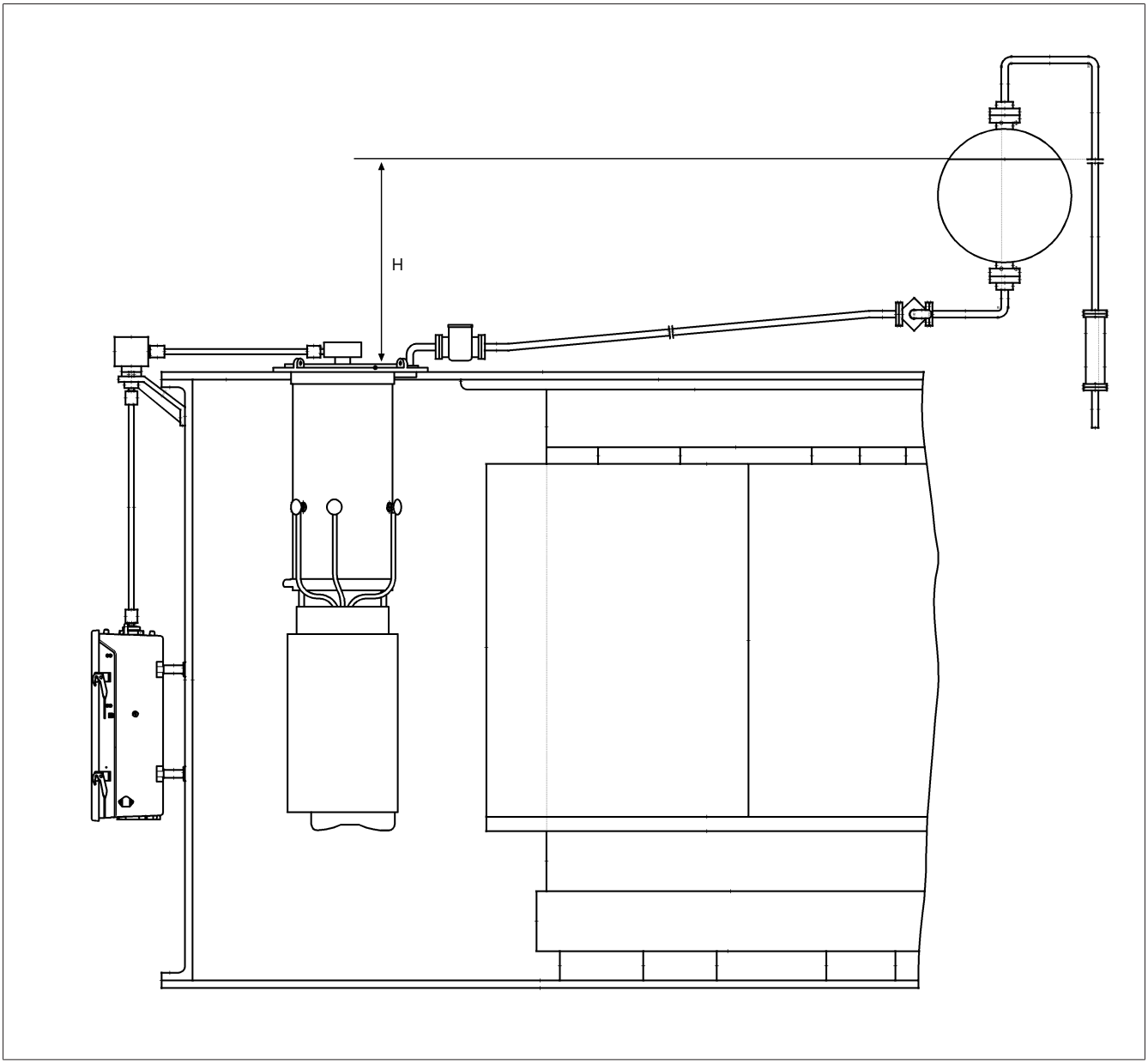
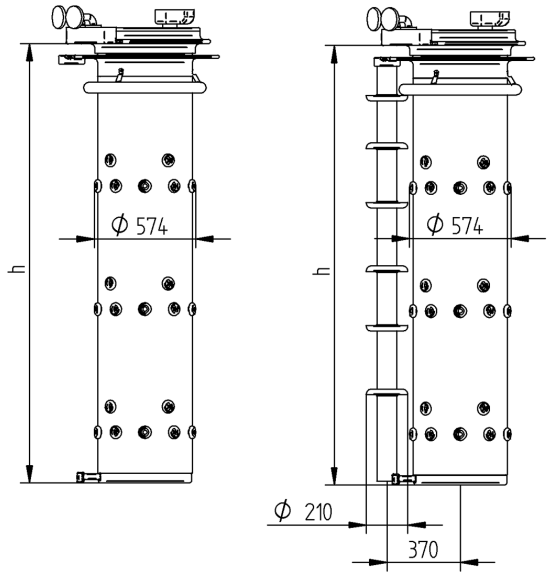


Figure 2: Transformer with on-load tap-changer, schematic presentation

1.3 On-load tap-changer models

without change-over selector

with change-over selector



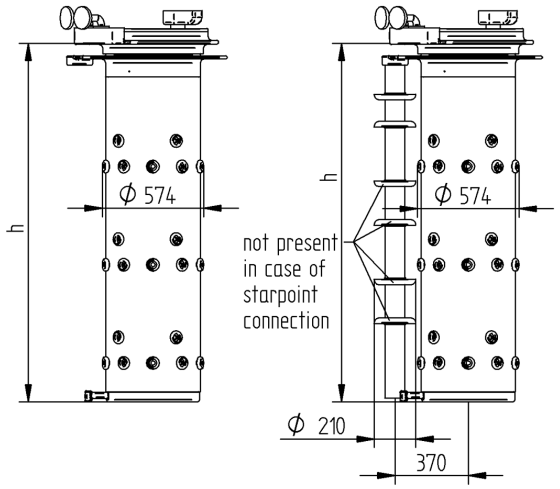
VV III 250/400/600 – D – 145 – 10/12 – O/W/G

installation length h in mm

U_m (kV)	without / with change-over selector
145	2224

without change-over selector

with change-over selector

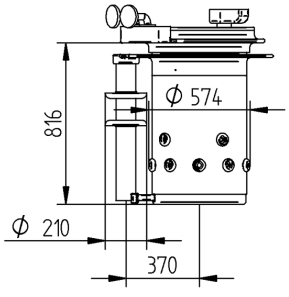


VV III 250/400/600 – Y/D – 40...76 – 10/12 – O/W/G

installation length h in mm

U_m (kV)	without / with change-over selector
40	1628
76	1810

VV I 401 – 76 – 10/12 – W/G



VV I 401 – 145 – 10/12 – W/G

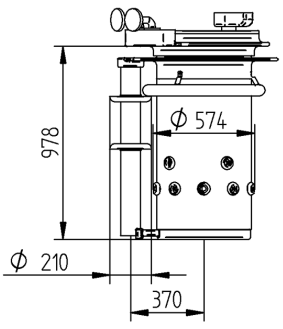


Figure 3: VACUTAP® VV

1.4 Overview of the basic connection diagrams

Overview of basic connections with designation of selector connection contacts in accordance with MR standard. This contact designation corresponds to the specifications in the on-load tap-changer dimensional drawings.

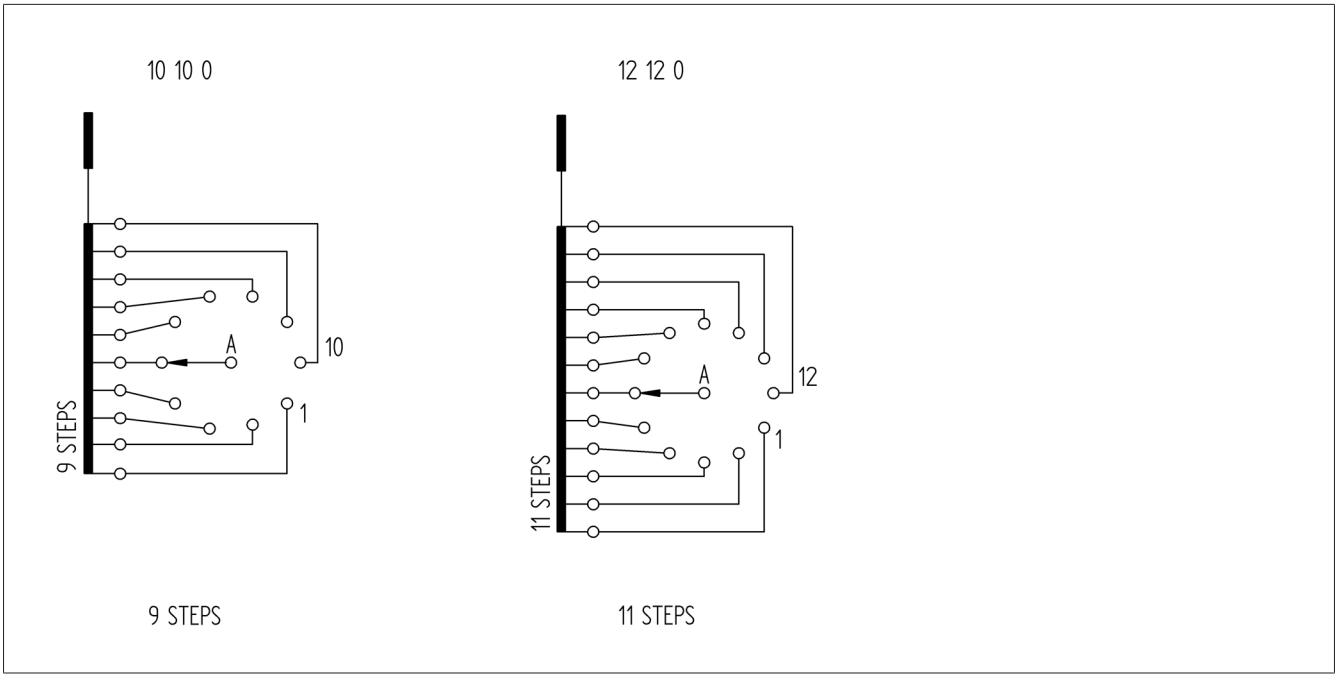


Figure 4: Connection without change-over selector

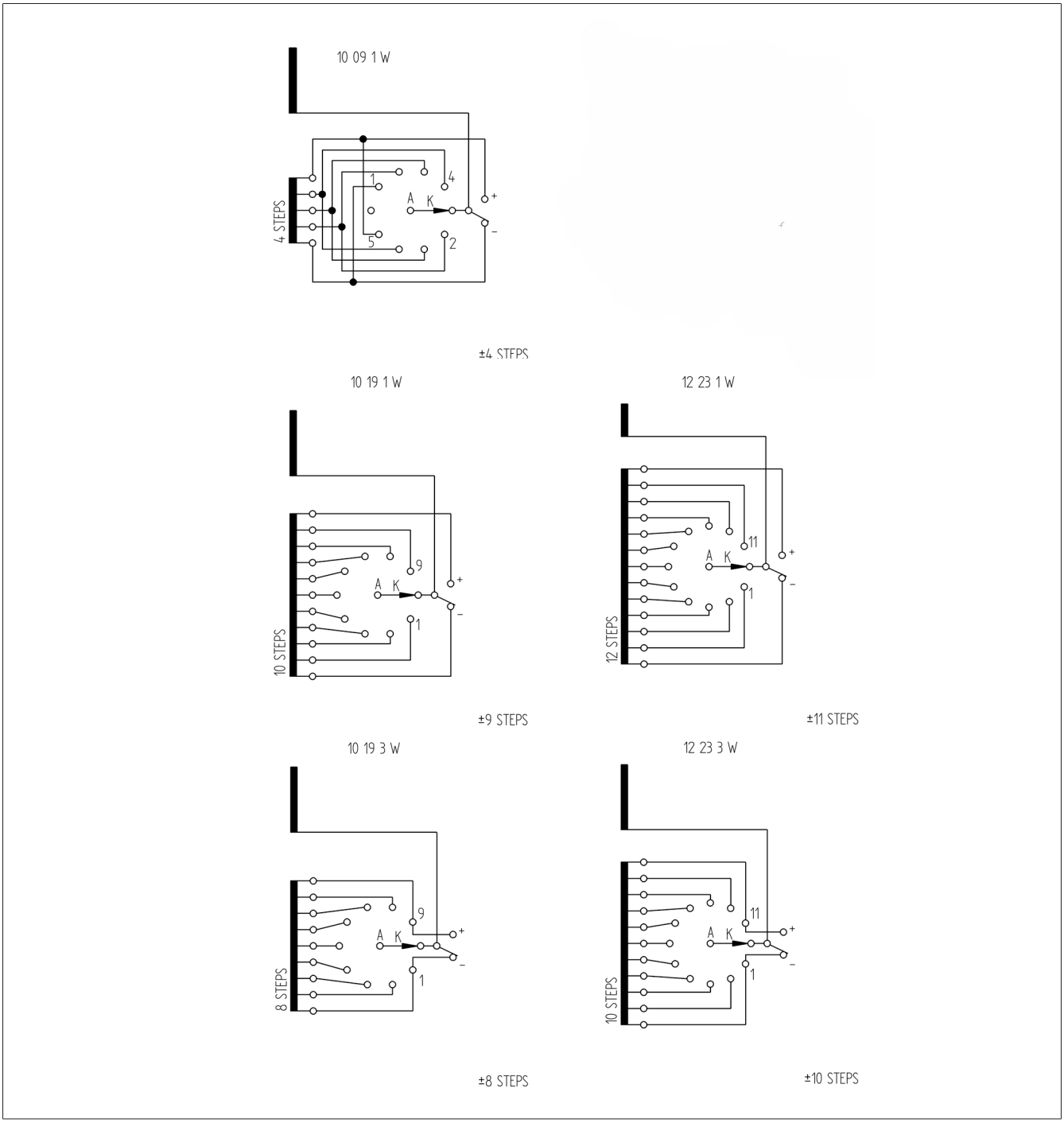


Figure 5: Reversing change-over selector connection

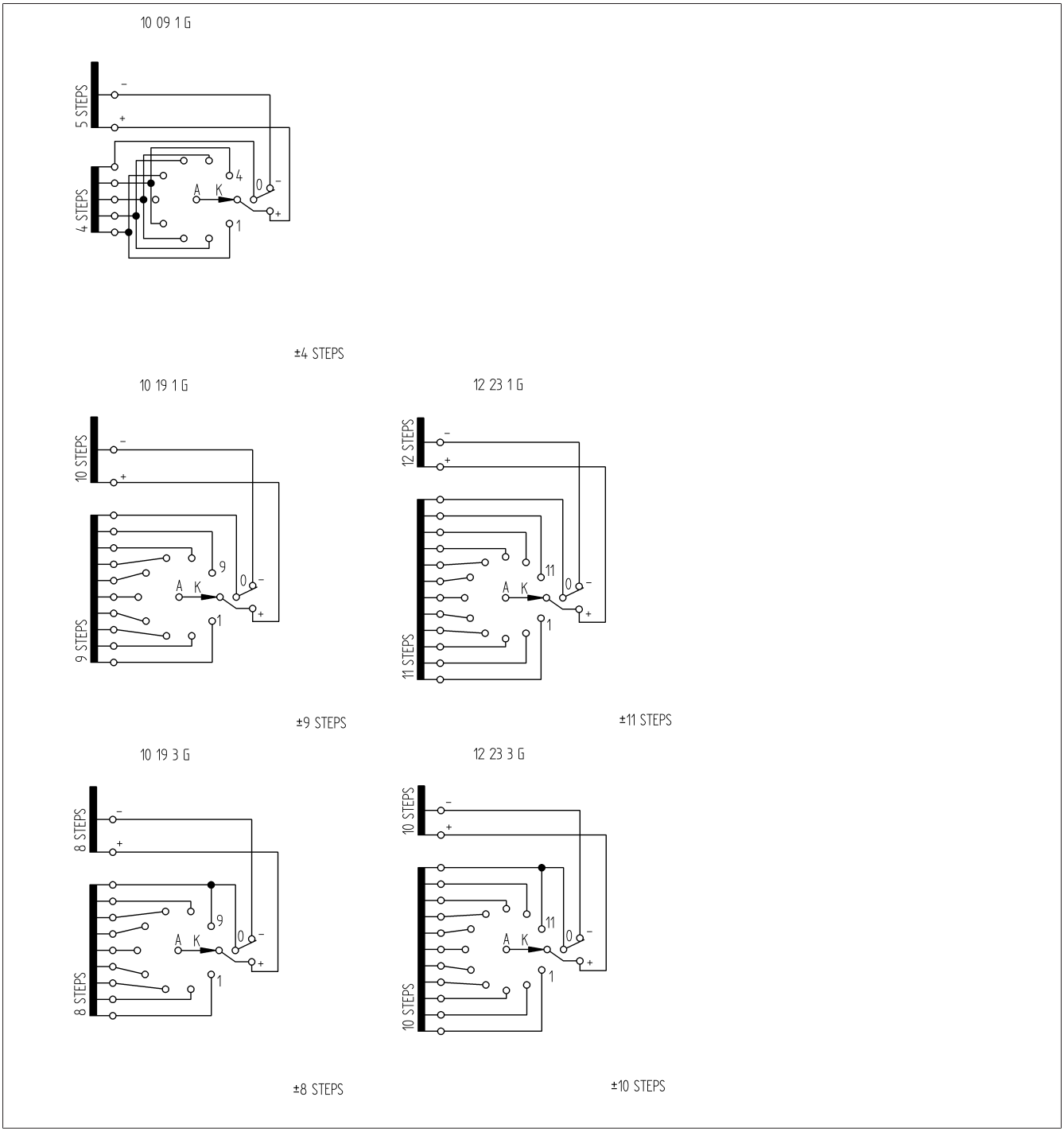


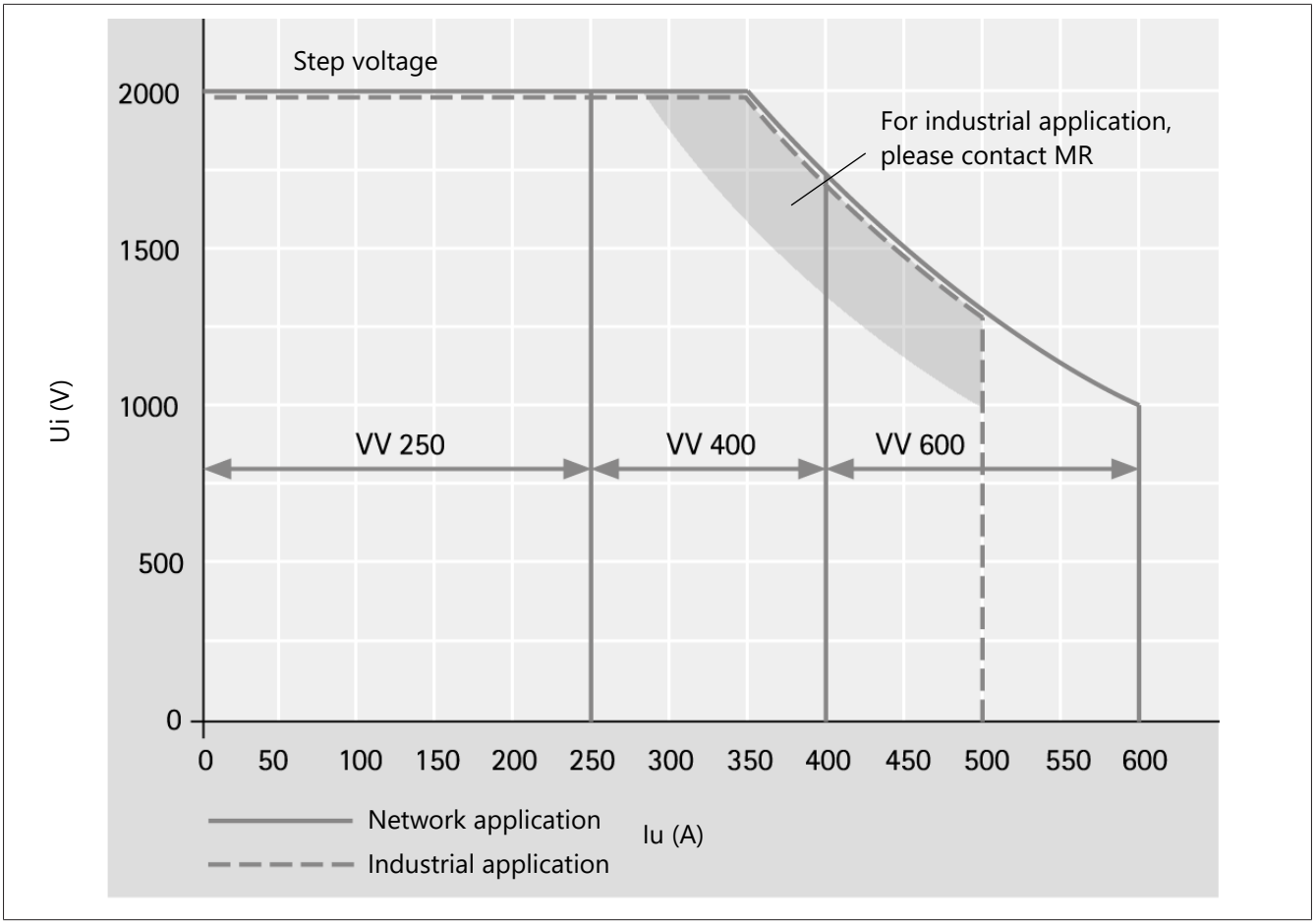
Figure 6: Coarse tap selector connection

2 Technical data

2.1 Rated through-current (Iu), rated step voltage (Ui) and step capacity (PSt)

	VV III 250	VV III 400/ VV I 401	VV III 600
Pitch of tap selector contact circle	10/12	10/12	10/12
I_u [A]	250	400	600
U_i [V]	2 000	2 000...1 700[1)]	2 000...1 000[1)]
P_{St} [kVA]	See P_{st}/I_u diagram		
[1)] see also U_i/I_u diagram			

Table 3: Rated through-currents (I_u), related rated step voltages (U_i) and step capacities (P_{St}), at highest and lowest rated step voltage of the step capacity curve (see following figure)



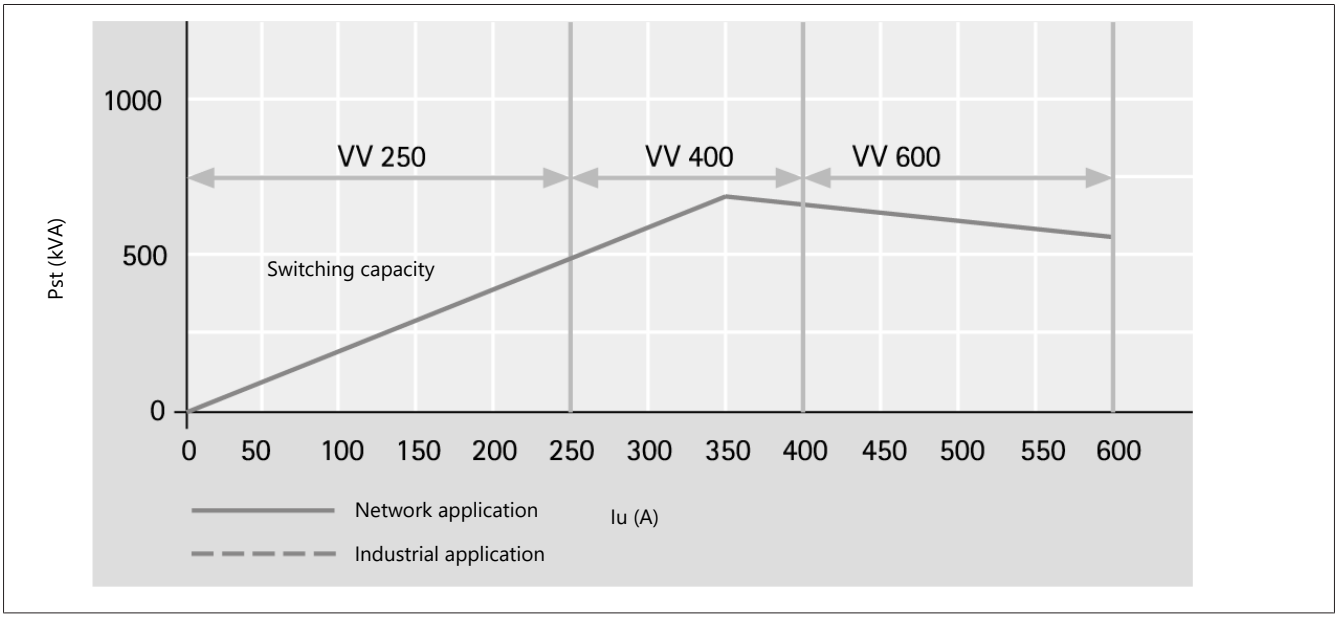


Figure 7: Step capacity (I_u, U_i)

2.2 Life of the vacuum interrupters

The vacuum interrupters should be replaced as a precautionary measure after 600,000 tap-change operations.

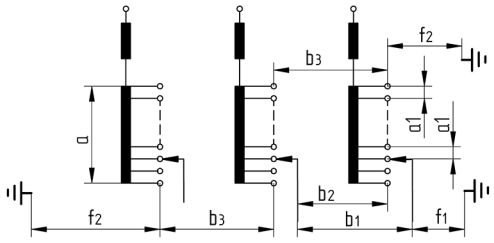
2.3 Rated withstand voltage of inner insulation

Insulation distances		Voltage waveform Duration	VV III xxx D	VV III xxx Y	VV I	
a	U_m = 40 kV U_m = 76 kV	kV 1.2/50 µs	200	200	U_m = 76 kV	200
		kV 50 Hz 1 min	50	50		50
	U_m = 123 kV	kV 1.2/50 µs		200		
		kV 50 Hz 1 min		50		
	U_m = 145 kV	kV 1.2/50 µs	250		U_m = 145 kV	200
		kV 50 Hz 1 min	65			65
a1	U_m = 40 kV U_m = 76 kV	kV 1.2/50 µs	100	100	U_m = 76 kV	100
		kV 50 Hz 1 min	20	20		20
	U_m = 123 kV	kV 1.2/50 µs		100		
		kV 50 Hz 1 min		20		
	U_m = 145 kV	kV 1.2/50 µs	130		U_m = 145 kV	130
		kV 50 Hz 1 min	30			30

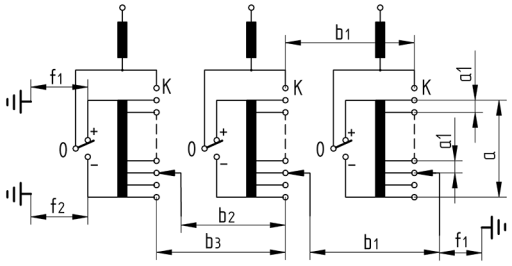
Insulation distances		Voltage waveform Duration	VV III xxx D			VV III xxx Y	VV I		
b			b_1	b_2	b_3				
	U_m = 40 kV	kV 1.2/50 µs	200	250	300	200			
		kV 50 Hz 1 min	70	80	90	85			
	U_m = 76 kV	kV 1.2/50 µs	350	490	520	200			
		kV 50 Hz 1 min	140	165	180	85			
	U_m = 123 kV	kV 1.2/50 µs				200			
		kV 50 Hz 1 min				85			
	U_m = 145 kV	kV 1.2/50 µs	650	730	800				
		kV 50 Hz 1 min	275	285	300				
c1	U_m = 40 kV U_m = 76 kV	kV 1.2/50 µs	400			400	U_m = 76 kV	400	
		kV 50 Hz 1 min	100			100		100	
	U_m = 123 kV	kV 1.2/50 µs				400			
		kV 50 Hz 1 min				100			
	U_m = 145 kV	kV 1.2/50 µs	500				U_m = 145 kV	500	
		kV 50 Hz 1 min	130					130	
c2			$c2_1$		$c2_2$				
	U_m = 40 kV	kV 1.2/50 µs	250		300	350			
		kV 50 Hz 1 min	80		90	140			
	U_m = 76 kV	kV 1.2/50 µs	490		520	350			
		kV 50 Hz 1 min	165		180	140			
	U_m = 123 kV	kV 1.2/50 µs				350			
		kV 50 Hz 1 min				140			
	U_m = 145 kV	kV 1.2/50 µs	730		800				
		kV 50 Hz 1 min	285		300				
f			f_1		f_2			f_1	f_2
	U_m = 40 kV	kV 1.2/50 µs	200		250	200			
		kV 50 Hz 1 min	70		80	70			
	U_m = 76 kV	kV 1.2/50 µs	350		490	350	U_m = 76 kV	350	490
		kV 50 Hz 1 min	140		165	140		140	165
	U_m = 123 kV	kV 1.2/50 µs				550			
		kV 50 Hz 1 min				230			
	U_m = 145 kV	kV 1.2/50 µs	650		730		U_m = 145 kV	650	730
		kV 50 Hz 1 min	275		285			275	285

Table 4: Rated withstand voltage (in kV) of the insulation distances (rated lightning impulse withstand voltage, rated short-duration power frequency withstand voltage) on the on-load tap-changer

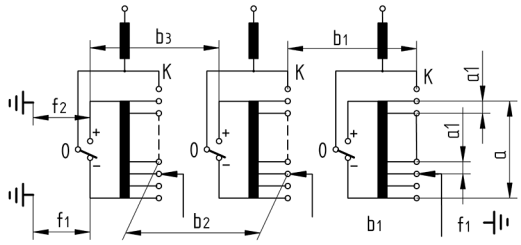
Without change-over selector



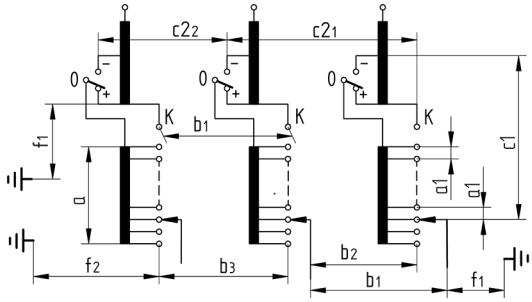
With reversing change-over selector in position +



With reversing change-over selector in position -



With coarse change-over selector in position +



With coarse change-over selector in position -

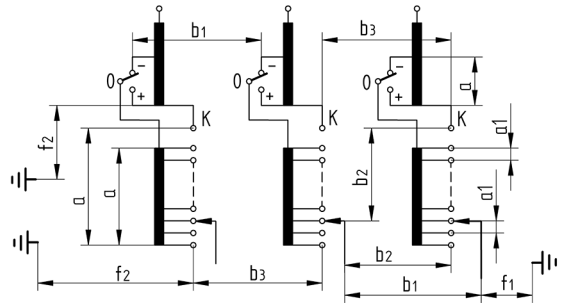


Figure 8: Specific voltage stress of the tap winding

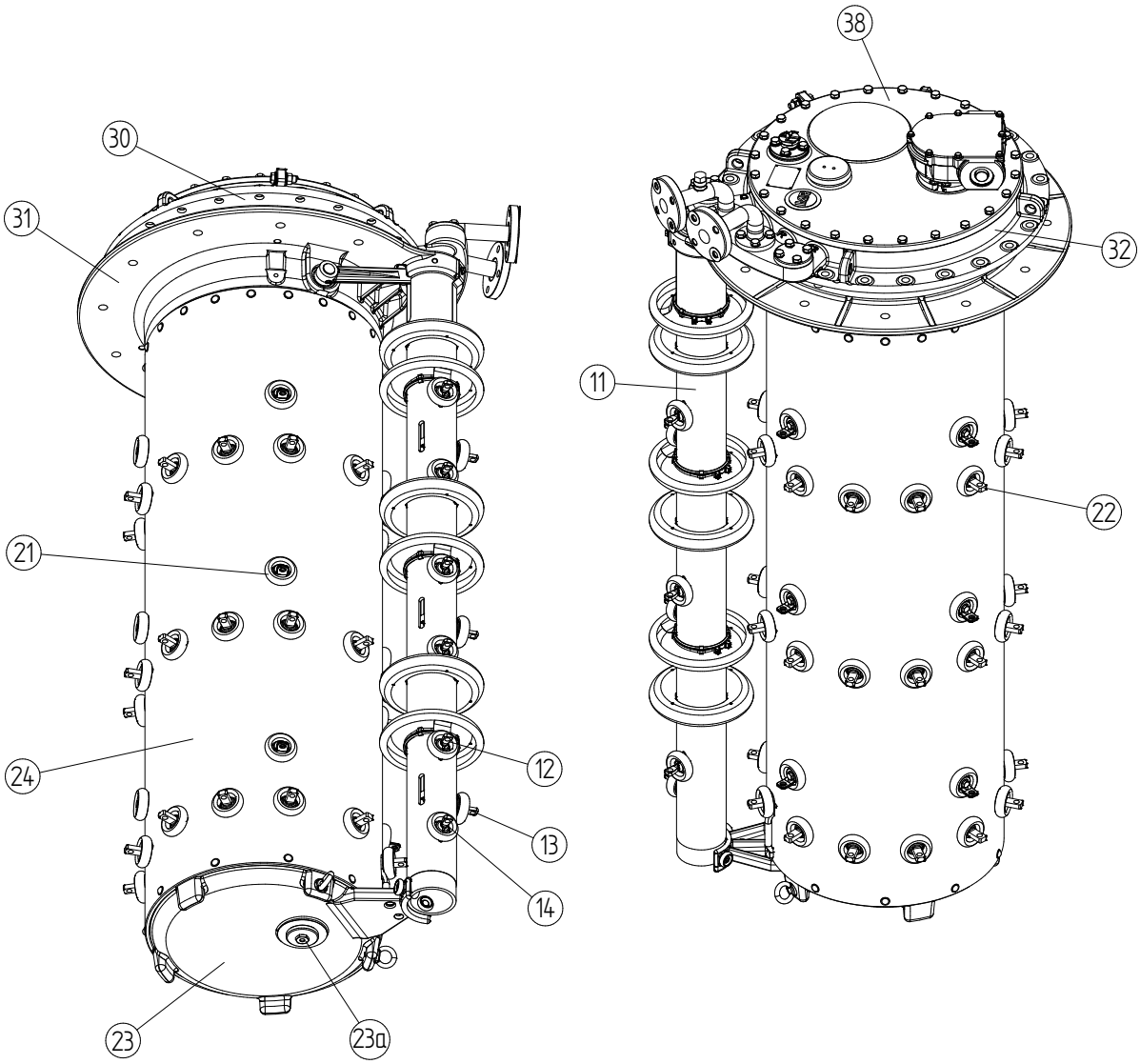
a1	= between tap selector contacts of the winding of one tap position (connected or not connected)
a	= between beginning and end of a tap winding and, for model with a coarse winding, also between the beginning and end of a coarse winding Note for coarse tap selector connection in the (-) change-over selector position: When stressed with impulse voltage, the permissible withstand voltage "a" between the end of a coarse winding connected with the K tap selector contact and the tap selector contact at the end of the tap winding of the same phase must be adhered to.
b	= between the fine tap selector contacts of different phases and between change-over selector contacts of different phases, which are connected with the start/end of a tap winding or with a tap selector contact. The permissible voltages between the contacts for tap-changers for delta connections depend on the positions of the change-over selector and tap selector. The different permissible withstand voltages b1, b2 or b3 must therefore be noted for an insulation distance: b_1 = between selected contacts for different phases b_2 = from the selected contact of one phase to non-selected contacts of other phases b_3 = between non-selected contacts for different phases

f	= between tap selector contacts and (+) change-over selector contacts to ground Different values apply for the delta connection version: f_1 = between take-off lead and ground and in (+) position of change-over selector between (+) change-over selector contacts and ground f_2 = between non-selected tap selector contacts and in (-) position of change-over selector between (+) contacts and ground
Additionally for coarse tap selector connection in (+) position of the change-over selector:	
c1	= from one (-) change-over selector contact to take-off lead of the same phase
c2	= between (-) change-over selector contacts of different phases and from one (-) change-over selector contact to one (+) change-over selector contact of another phase For tap-changers, different permissible values should be noted for the delta connection: $c2_1$ = between a (-) change-over selector contact and a (+) change-over selector contact of another phase $c2_2$ = between (-) change-over selector contacts of different phases

3 Drawings

3.1 Overview drawings

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(11) = Change-over selector pipe

(12) = Change-over selector terminal "+"

(13) = Change-over selector terminal "0"

(14) = Change-over selector terminal "–"

(21) = Take-off terminal

(22) = Fine tap selector terminal

(23) = Oil compartment bottom with kerosene drain plug 23a

(24) = Tap changer oil compartment

(30) = Tap changer head complete with 31 und 32 , refer to 898 863 : and 737 060 :

(31) = Tap changer head – bottom part

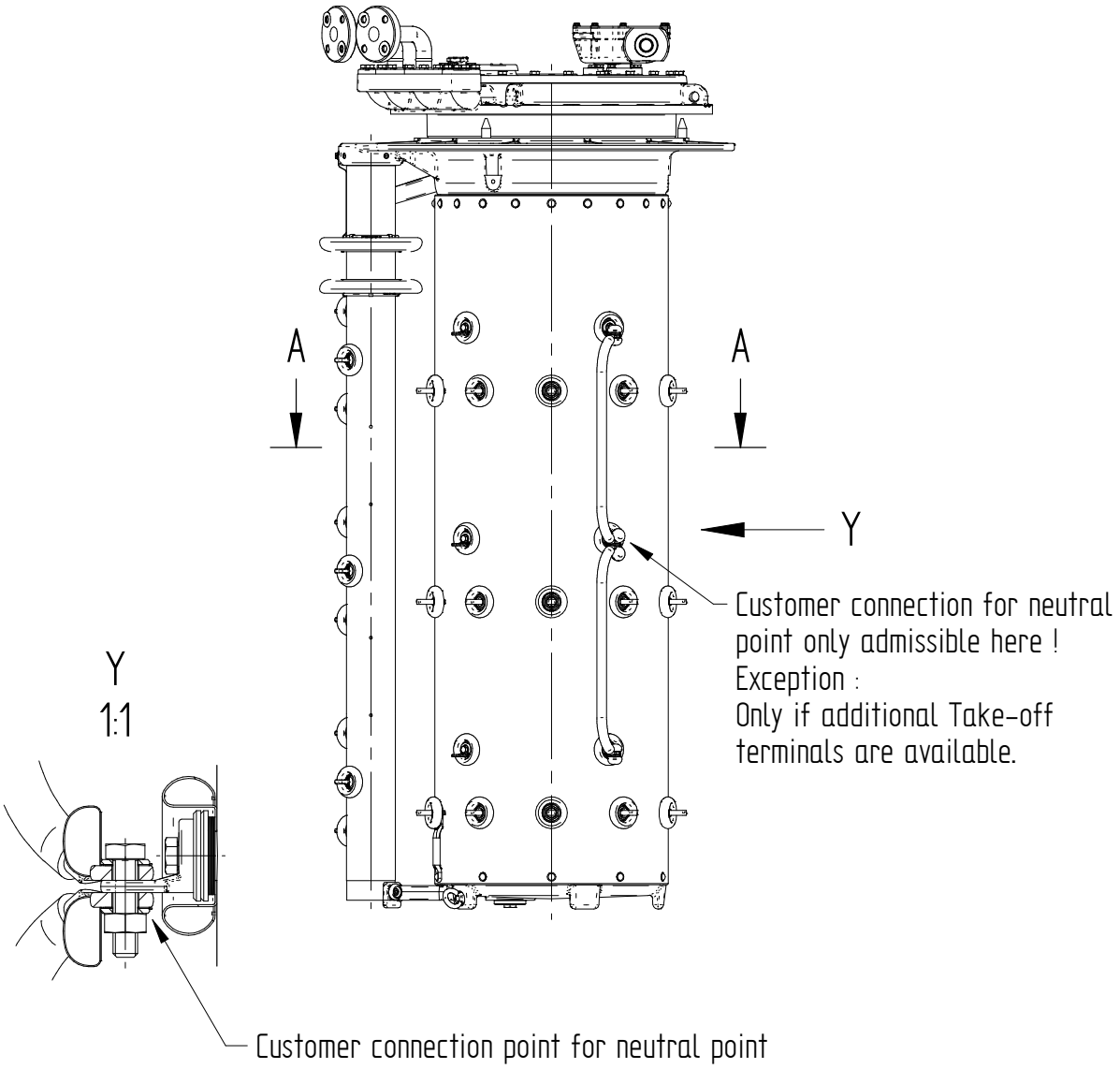
(32) = Tap changer head – top part

(38) = Tap changer head cover

	On-Load Tap-Changer VACUTAP ® VV Classification	737774 0E

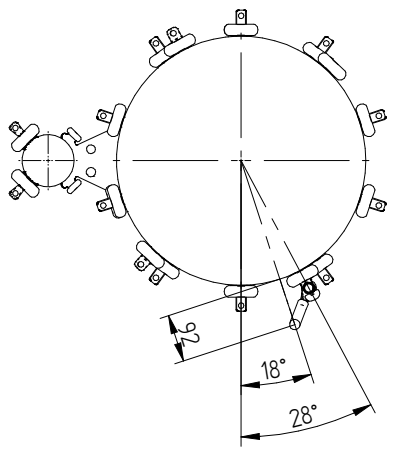
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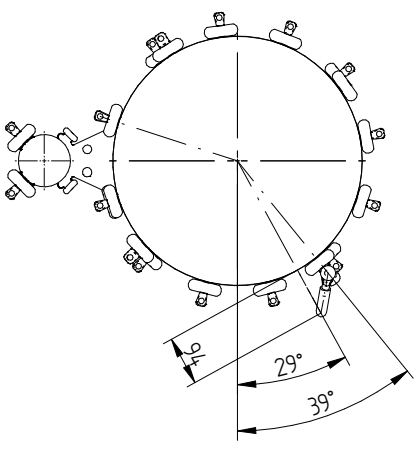


Position of neutral point connection lead

A–A , 10 pitch



A–A , 12 pitch

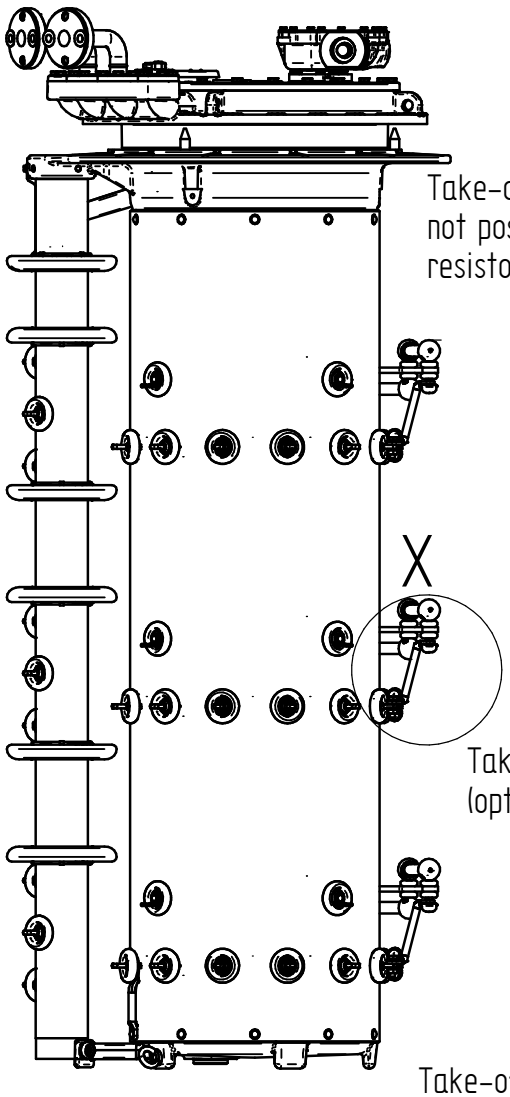


The position of the neutral point connection lead must not be changed !

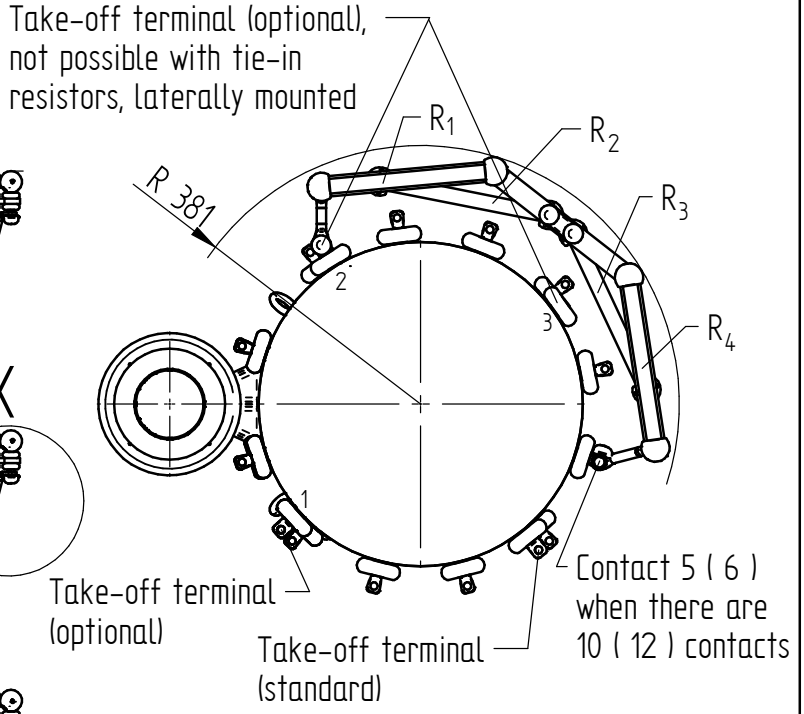
MR	On-Load Tap-Changer VACUTAP ® V V Neutral point connection lead	738806 0E

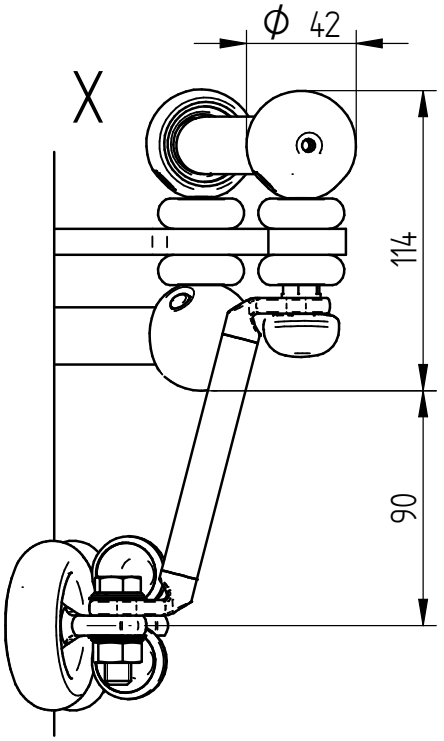
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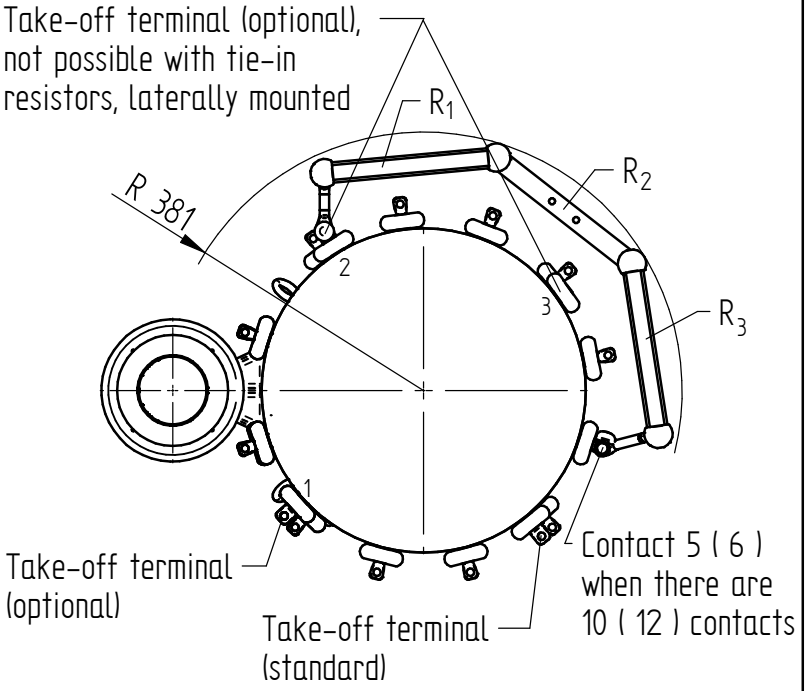


4 tie-in resistors per phase





3 tie-in resistors per phase



On-Load Tap-Changer V V – 76 kV – in 12 pitch design

3E	052508	21.12.04	DC				
2E	046759	11.08.03	DC				
1E	042782	21.11.02	DC				
NEU	035017	11.07.01	DC				
Ae. I	Aend.-Nr.	Tag	Name				

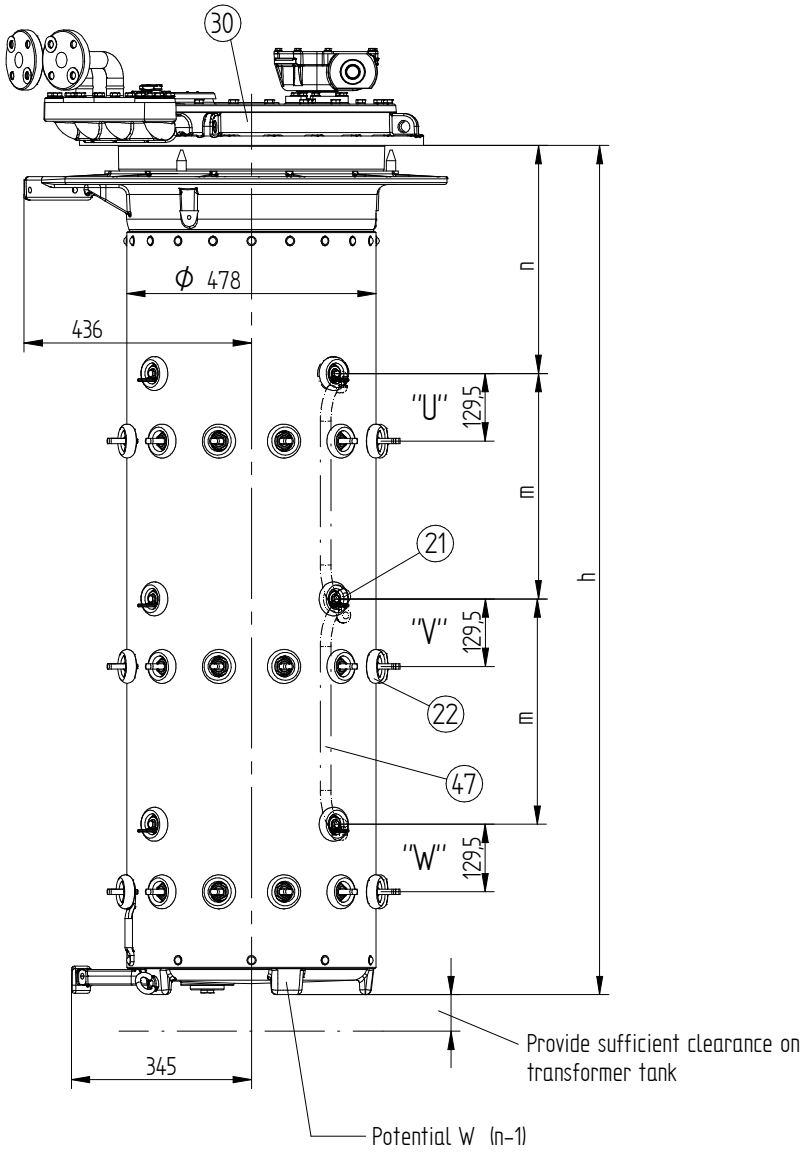
gez. DC 21.12.04
gepr. Albrecht

MR	On-Load Tap-Changer VACUTAP® V V Tie-in resistors attached laterally	SED 1040597-000-03 8994073E

DIN A3

3.2 Dimensional drawings

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(21) = Take-off terminal
(22) = Fine tap selector terminal
Dimensions refer to 737 776 :

(30) = Tap changer head , dimensions refer to 898 863 : and 737 060 :

(47) V V – Y only : Neutral point connection lead , dimensions refer to 738 806 :

U_m [kV]		40	76
Dimensions [mm]	m	432	497
	n	437	489
	h	1628	1810

⚠ The connecting diagram is binding for the designation of the terminals and phases

	SE	1002693	02.10.09	SPL
	SE	1010449	20.01.09	SPL
	ES-TR	Change-No.	Date	Name
diffr. SPL 20.01.09				
chkd				

MR

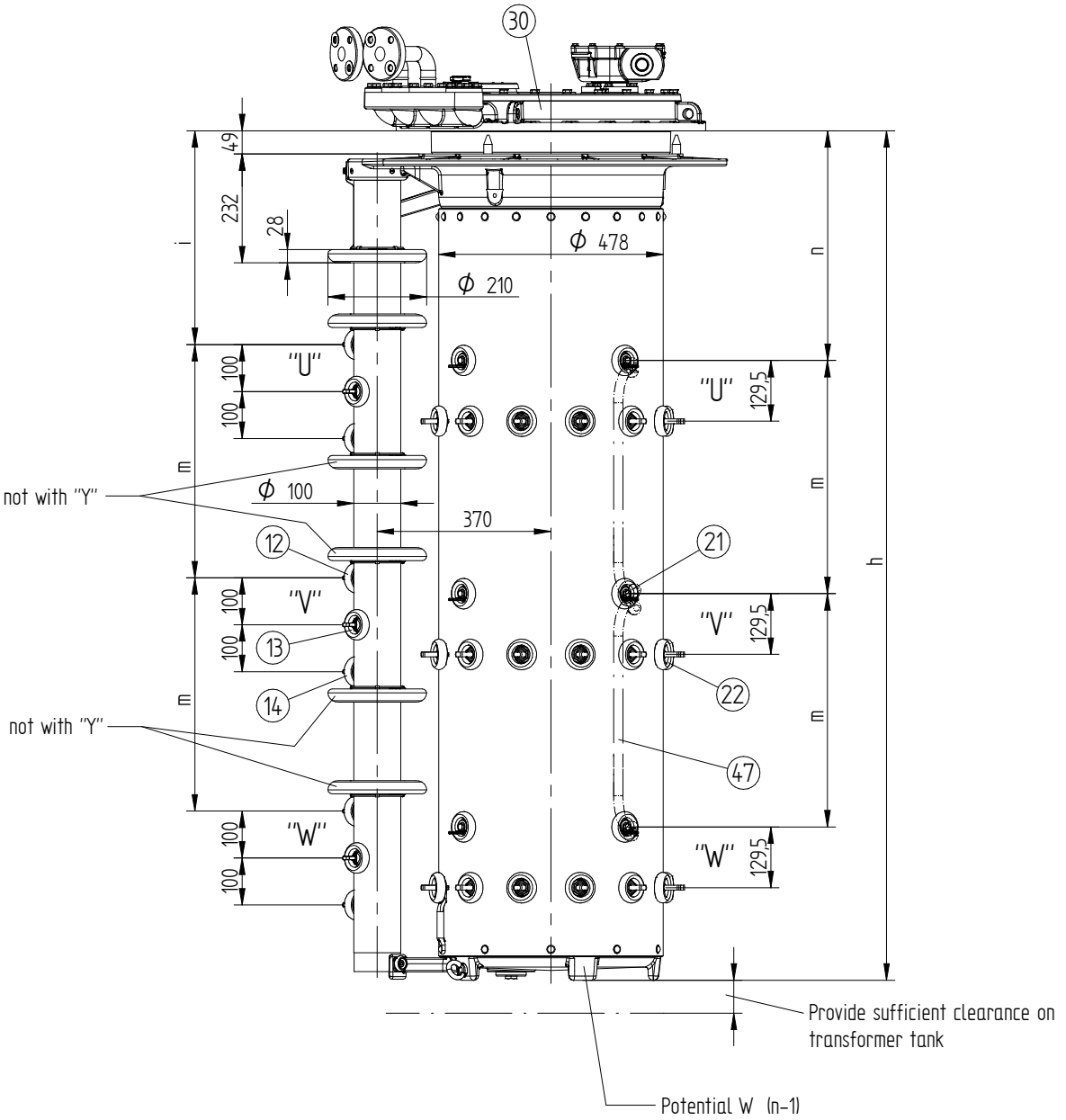
On-Load Tap-Changer VACUTAP ® V V

V V III 250/400/600-Y/D-40...76-10/12-0

899120 5E

DIN A1

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(12) = Change-over selector terminal "+"
(13) = Change-over selector terminal "0"
(14) = Change-over selector terminal "-"
(21) = Take-off terminal
(22) = Fine tap selector terminal

Dimensions refer to 737 775 :

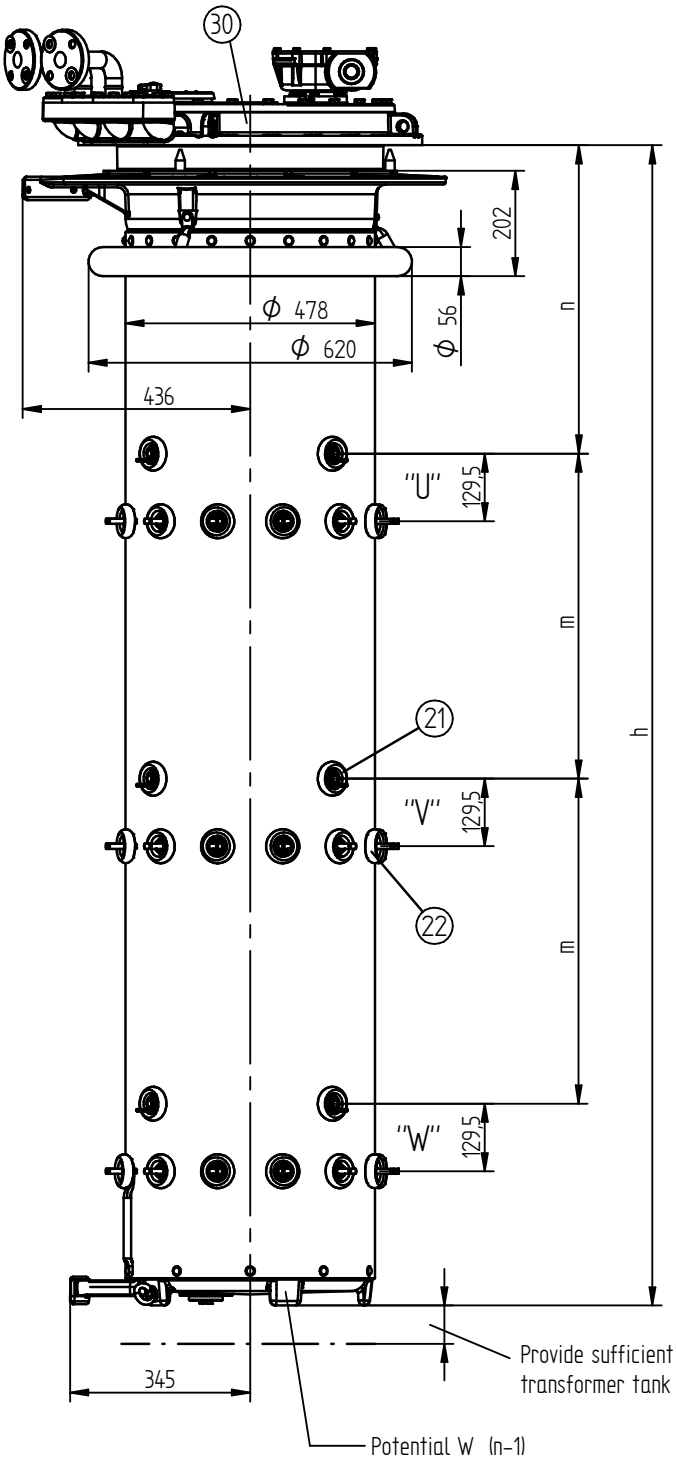
U_m [kV]		40	76
Dimensions [mm]	i	404	456
	m	432	497
	n	437	489
	h	1628	1810

(30) = Tap changer head , dimensions refer to 898 863 : and 737 060 :

(47) V V – Y only : Neutral point connection lead , dimensions refer to 738 806 :

⚠ The connecting diagram is binding for the designation of the terminals and phases

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(21) = Take-off terminal
(22) = Fine tap selector terminal
(30) = Tap changer head , dimensions refer to 898 863 : and 737 060 :

Dimensions refer to 737 776 :

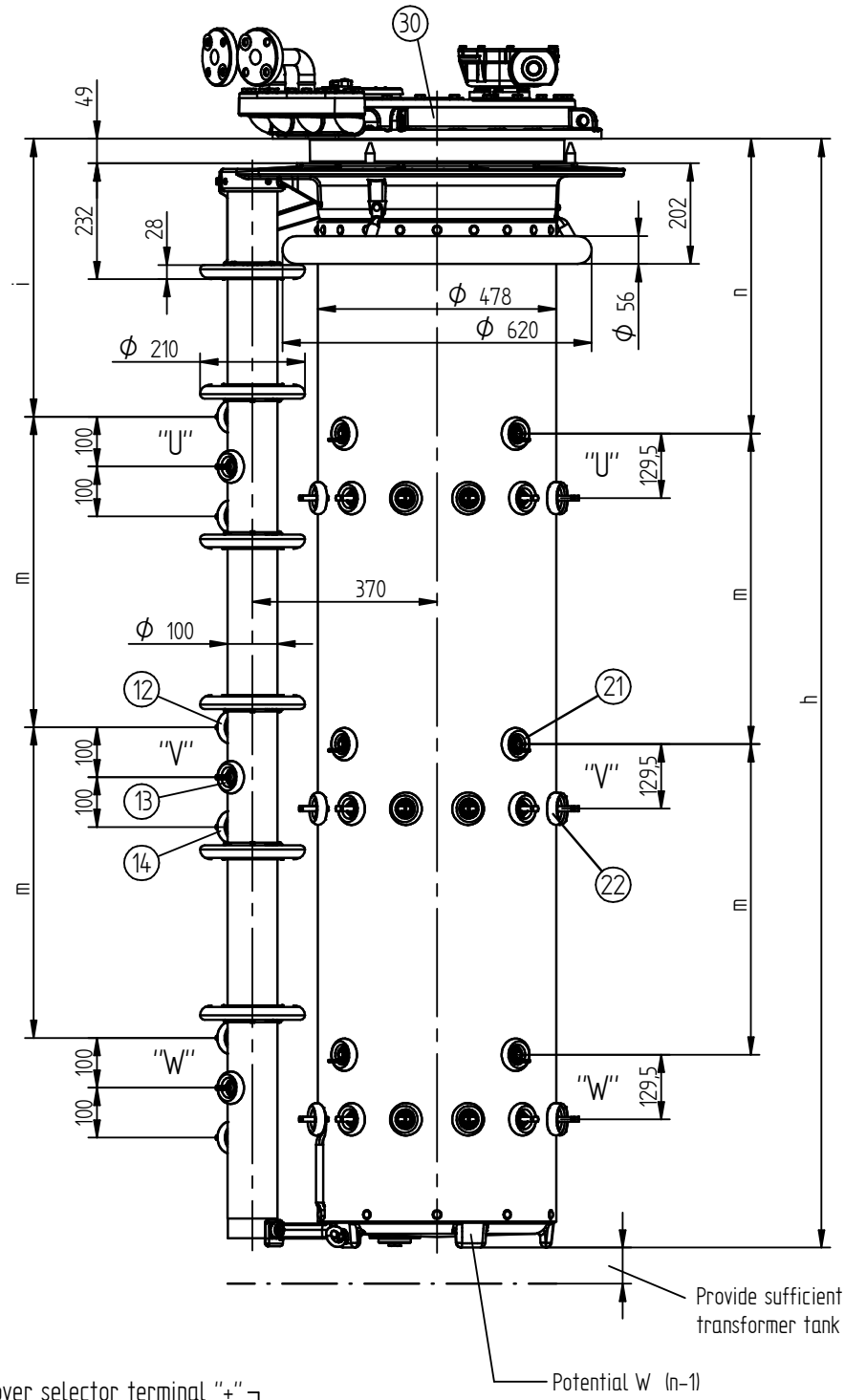
U_m [kV]		145
Dimensions [mm]	m	623
	n	591
	h	2224

The connecting diagram is binding for the designation of the terminals and phases

MR	On-Load Tap-Changer VACUTAP ® V V V V III 250/400/600-D-145-10/12-0	899163 5E

DIN A1

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(12) = Change-over selector terminal "+"
(13) = Change-over selector terminal "0"
(14) = Change-over selector terminal "-"
(21) = Take-off terminal
(22) = Fine tap selector terminal

Dimensions refer to 737 775 :

(30) = Tap changer head , dimensions refer to 898 863 : and 737 060 :

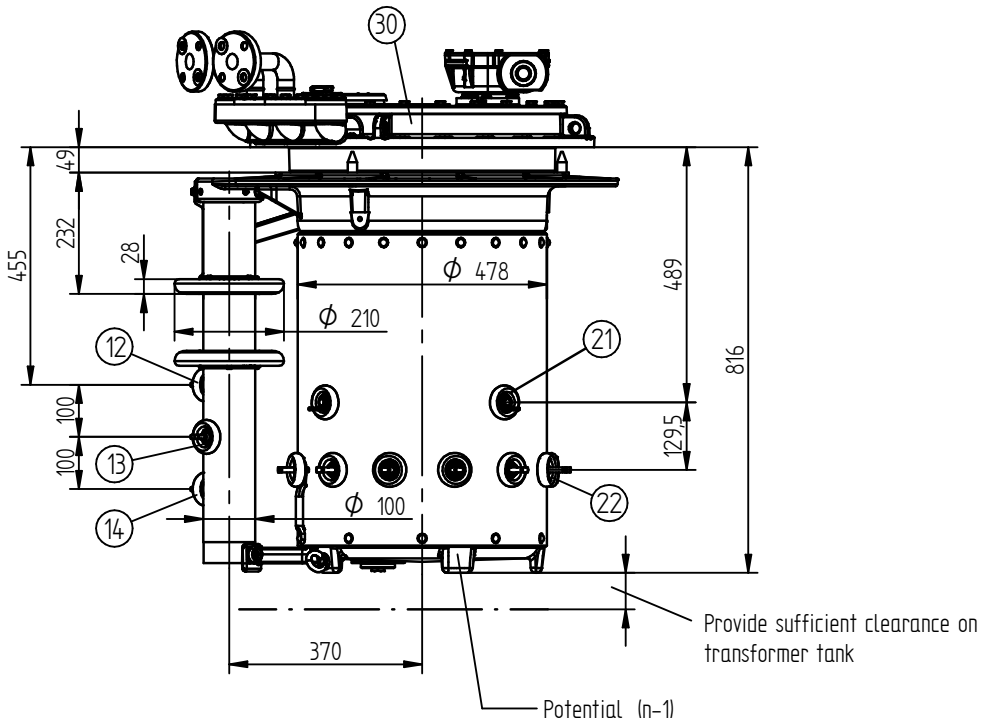
U_m [kV]		145
Dimensions [mm]	i	558
	m	623
	n	591
	h	2224

The connecting diagram is binding for the designation of the terminals and phases

	On-Load Tap-Changer VACUTAP ® V V V V III 250/400/600-D-145-10/12-W/G	899162 5E

DIN A1

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(12) = Change-over selector terminal "+"
(13) = Change-over selector terminal "0"
(14) = Change-over selector terminal "–"
(21) = Take-off terminal
(22) = Fine tap selector terminal

Dimensions refer to 737 775 :

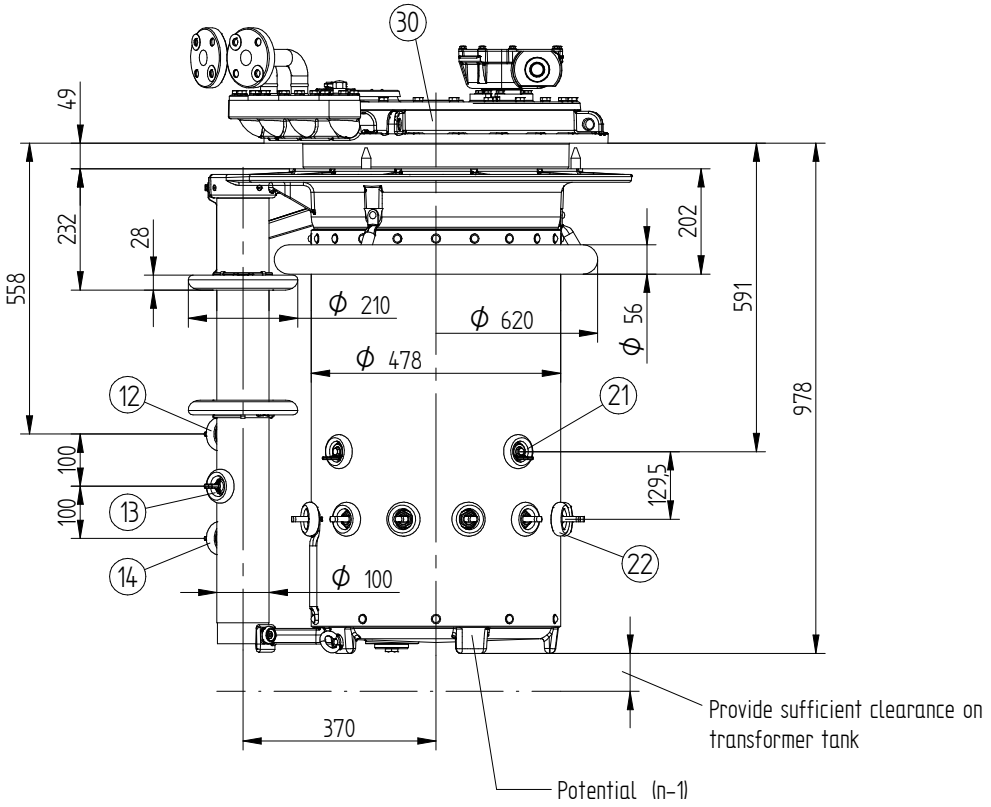
(30) = Tap changer head , dimensions refer to 898 863 : and 737 060 :

⚠ The connecting diagram is binding for the designation of the terminals and phases

	On-Load Tap-Changer VACUTAP ® V V V V I 401 – 76 – 10/12 – W/G	899121 5E

DIN A1

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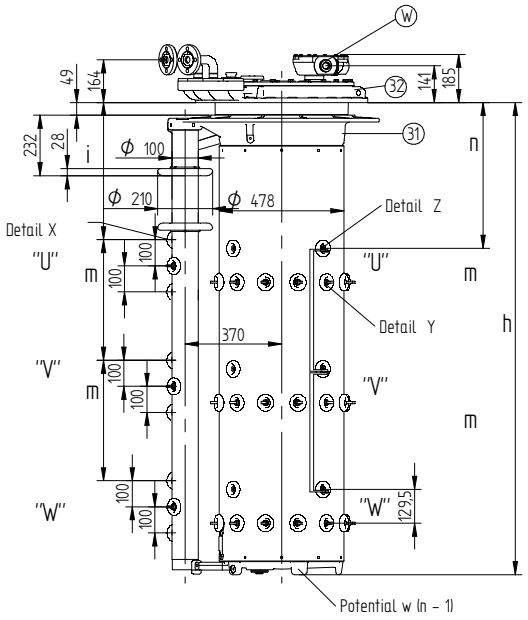
(12) = Change-over selector terminal "+"
(13) = Change-over selector terminal "0"
(14) = Change-over selector terminal "–"
(21) = Take-off terminal
(22) = Fine tap selector terminal

Dimensions refer to 737 775 :

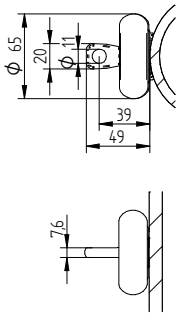
(30) = Tap changer head , dimensions refer to 898 863 : and 737 060 :

⚠ The connecting diagram is binding for the designation of the terminals and phases

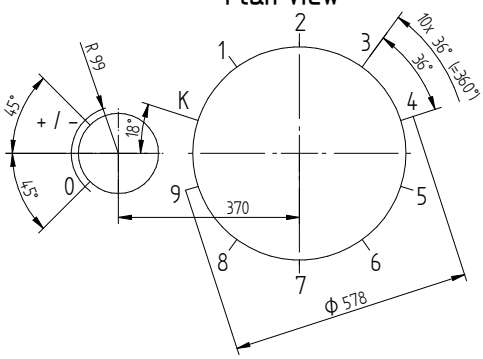
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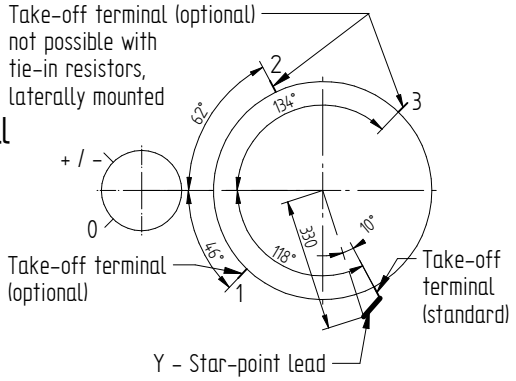
X
Change-over selector terminal



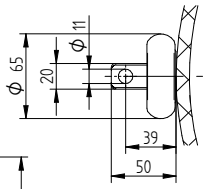
10 pitch contact arrangement
Plan view

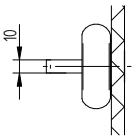


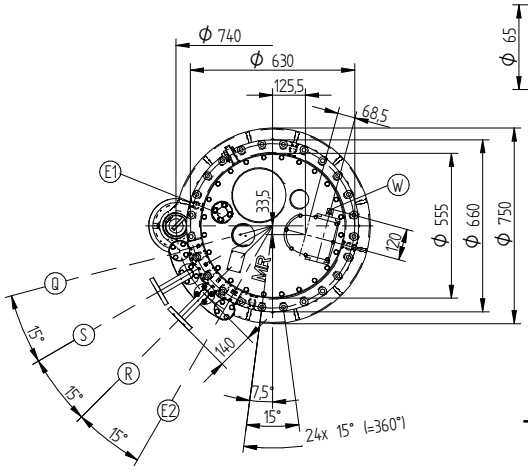
Location of 10 pitch take-off terminals
Plan view



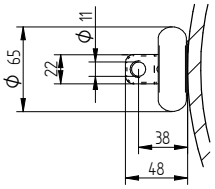
Y
Fine tap selector terminal

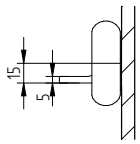




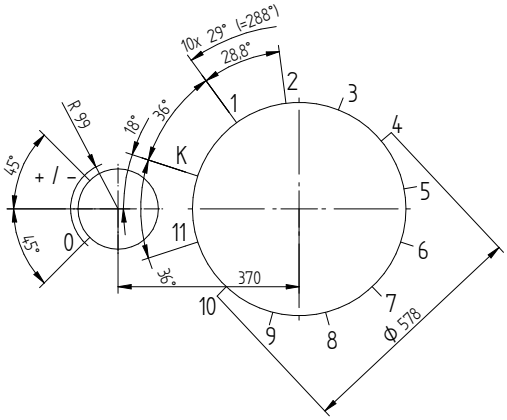


Z
Take-off terminal

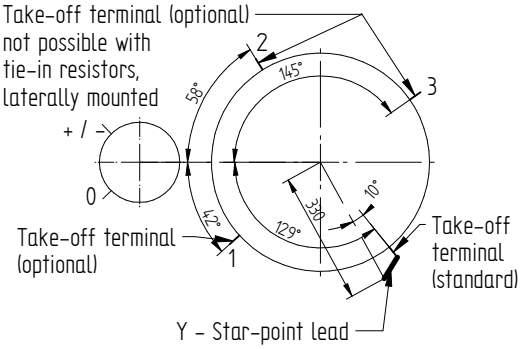




12 pitch contact arrangement
Plan view



Location of 12 pitch take-off terminals
Plan view



(E1) -Bleeding facility for Tap-Changer head

(E2) -Bleeding facility for space under the head outside the oil compartment

(Q) -Pipe connection (optional)

(R) -Connection for protective relay

(S) -Pipe connection

(W) -Drive shaft

(31) -Tap-Changer head - bottom part

(32) -Tap-Changer head - top part
(can be turned towards the bottom part of the head in steps of 15°)
the Tap-Changer head cover cannot be turned

U_m [kV]		123
Dimensions [mm]	i	526
	m	462
	n	559
	h	1810

The connection diagram is binding for the designation of the terminals and phases

	DATE	NAME	DOCUMENT NO.	
DFTR.	03.06.2024	STEPHAN	SED 1040859 000 04	
CHKD.	03.06.2024	STREMPEL	CHANGE NO.	SCALE 1:10
STAND.	04.06.2024	KLEYN	1129618	

DIMENSION IN mm EXCEPT AS NOTED

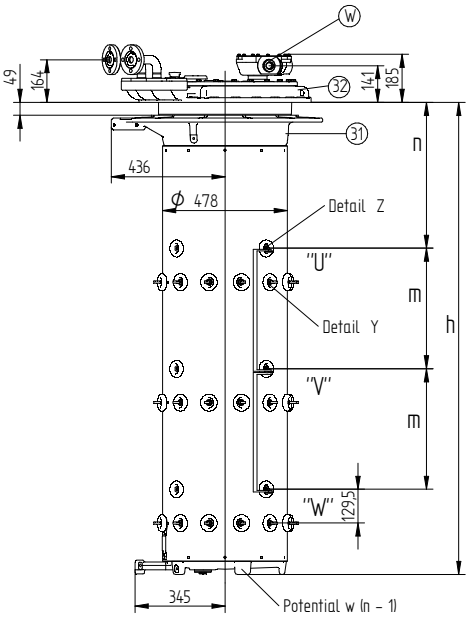
ON-LOAD TAP-CHANGER VACUTAP® VV®
VV III 250/400/600-Y-123-10/12-W/G

SERIAL NUMBER -

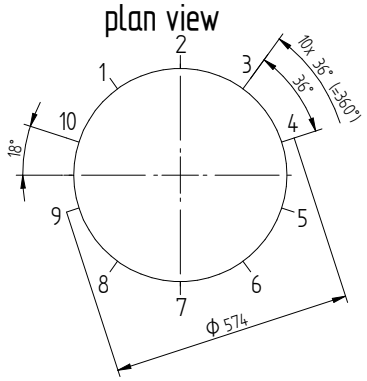
MATERIAL NUMBER 8996772E

SHEET 1/1

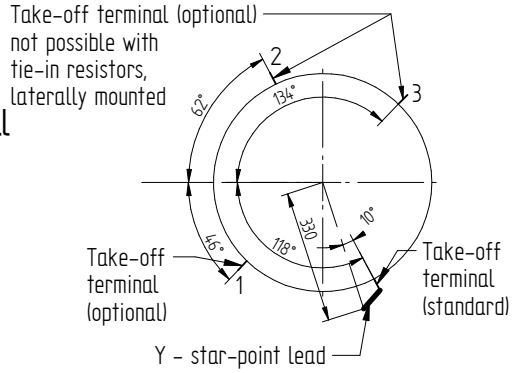
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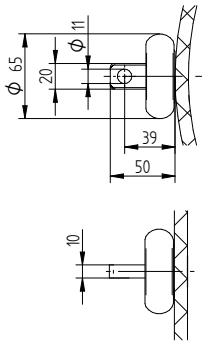
10 pitch contact arrangement plan view

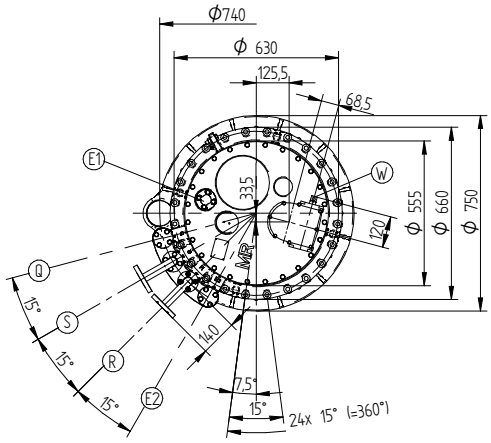


Location of 10 pich take-off terminals plan view

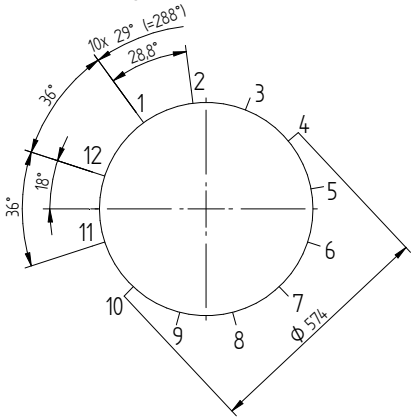


Y
Fine tap selector terminal

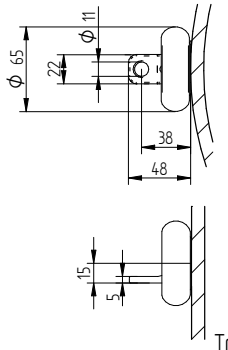




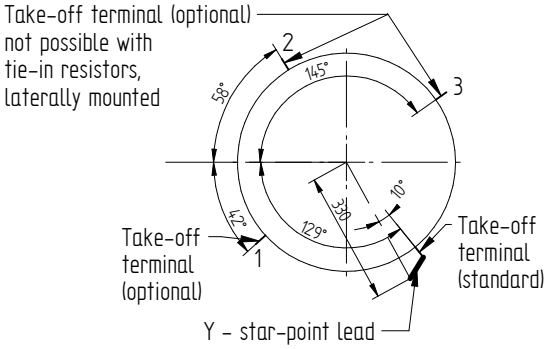
12 pitch contact arrangement plan view



Z
Take-off terminal



Location of 12 pitch take-off terminals Plan view



(E1) –Bleeding facility for Tap-Changer head

(E2) –Bleeding facility for space under the head outside the oil compartment

(Q) –Pipe connection (optional)

(R) –Connection for protective relay

(S) –Pipe connection

(W) –Drive shaft

(31) –Tap-Changer head – bottom part

(32) –Tap-Changer head – top part
(can be turned towards the bottom part of the head in steps of 15°)
the Tap-Changer head cover cannot be turned

U_m [kV]		123
Dimensions [mm]	m	462
	n	559
	h	1810

The connection diagram is binding for the designation of the terminals and phases

SCALE 1 : 10

	NAME	DATE
DRAWN	STEPHAN	03.06.2024
CHKD.	STREMPEL	03.06.2024
STAND.	KLEYN	04.06.2024

DOCUMENT NO. SED 1993305 000 02
CHANGE NO. 1129618
SCALE 1:10

DIMENSION IN mm EXCEPT AS NOTED

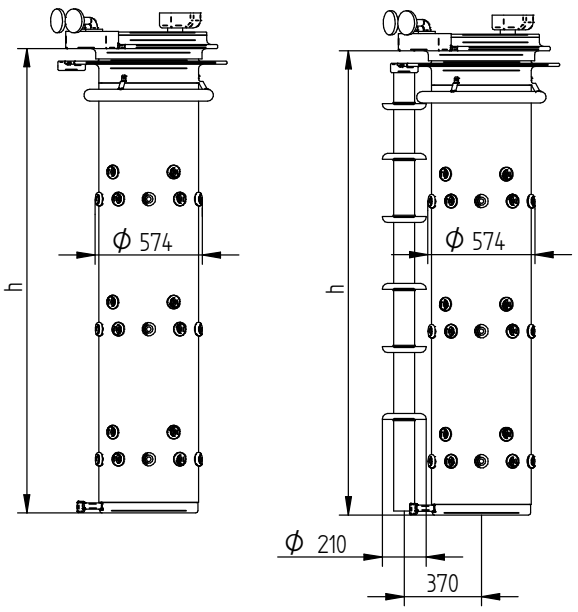
ON-LOAD TAP-CHANGER VACUTAP® VV®
VV III 250/400/600-Y-123-10/12-0

SERIAL NUMBER –
MATERIAL NUMBER 7380100E
SHEET 1/1

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without change-over selector

with change-over selector



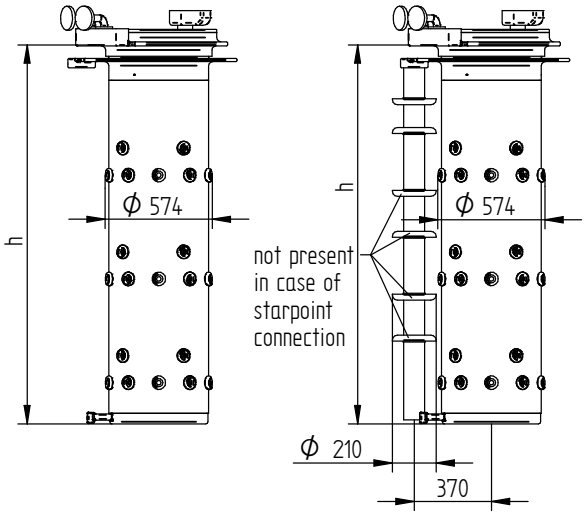
VV III 250/400/600 – D – 145 – 10/12 – 0/W/G

installation length h in mm

U_m (kV)	without / with change-over selector
145	2224

without change-over selector

with change-over selector

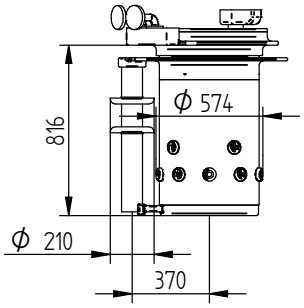


VV III 250/400/600 – Y/D – 40...76 – 10/12 – 0/W/G
VV III 250/400/600 – Y – 123 – 10/12 – 0/W/G

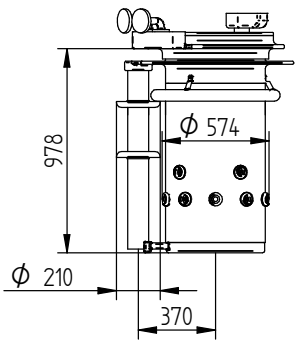
installation length h in mm

U_m (kV)	without / with change-over selector
40	1628
76	1810
123Y	

VV I 401 – 76 – 10/12 – W/G



VV I 401 – 145 – 10/12 – W/G



	DATE	NAME	DOCUMENT NO.	
DFTR.	04.06.2024	STEPHAN	SED 1040736 000 03	
CHKD.	04.06.2024	STREMPEL	CHANGE NO.	SCALE
STAND	04.06.2024	KLEYN	1129618	1:25

DIMENSION IN mm EXCEPT AS NOTED

ON-LOAD TAP-CHANGER VACUTAP® VV®
SURVEY OF VERSIONS-NUMBER OF POLES, CHANGE-OVER SELECTOR, INSTALLATION LENGTH

SERIAL NUMBER –

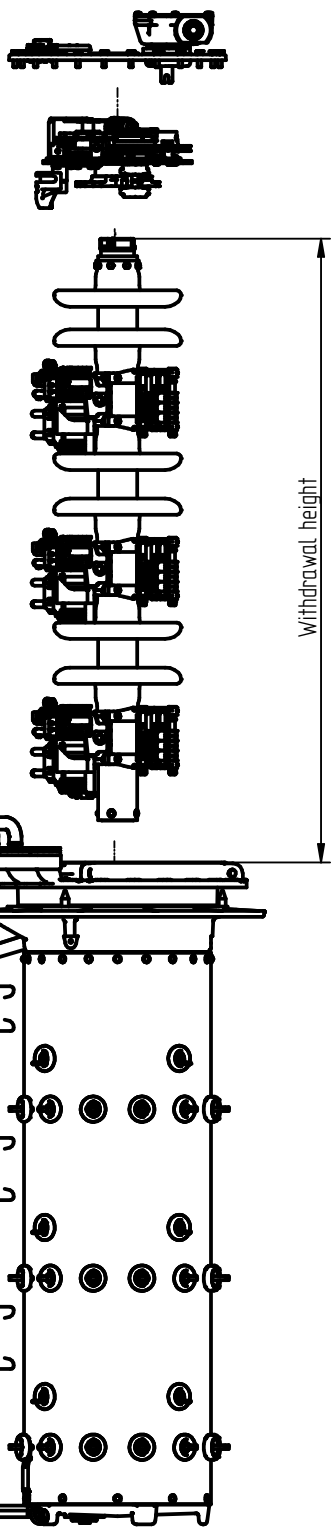
MATERIAL NUMBER 8995411E

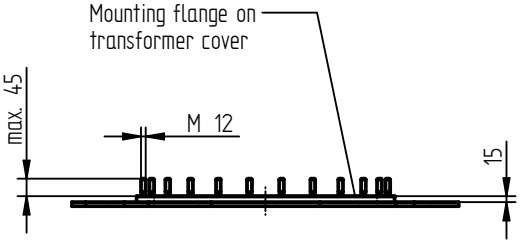
SHEET 1 / 1

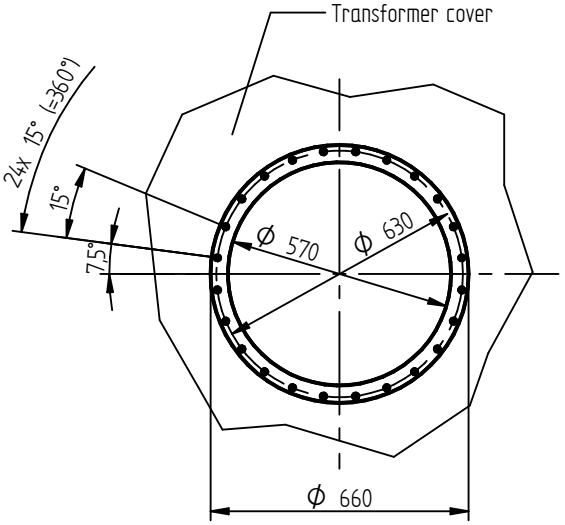
A3

3.3 Installation drawings

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On-Load Tap-Changer	Withdrawal height
V V I – 76 kV	920 mm
V V I – 145 kV	1070 mm
V V III – 40 kV V V III – 76 kV V V III – 123-Y kV	1910 mm
V V III – 145 kV	2310 mm

The withdrawal height corresponds to the clearance between the cover flange and the lifting device of the OLTC insert, while the mounting rod is inserted. Additional fixing pads, which may be necessary for affixing the OLTC insert, have to be considered too.

	DATE	NAME	DOCUMENT NO.	
DFTR.	04.06.2024	STEPHAN	SED 2058776 000 01	
EHKO.	04.06.2024	STREMPEL	CHANGE NO.	SCALE
STAND	04.06.2024	KLEYN	1129618	

DIMENSION IN mm EXCEPT AS NOTED

ON-LOAD TAP-CHANGER VACUTAP® VV®

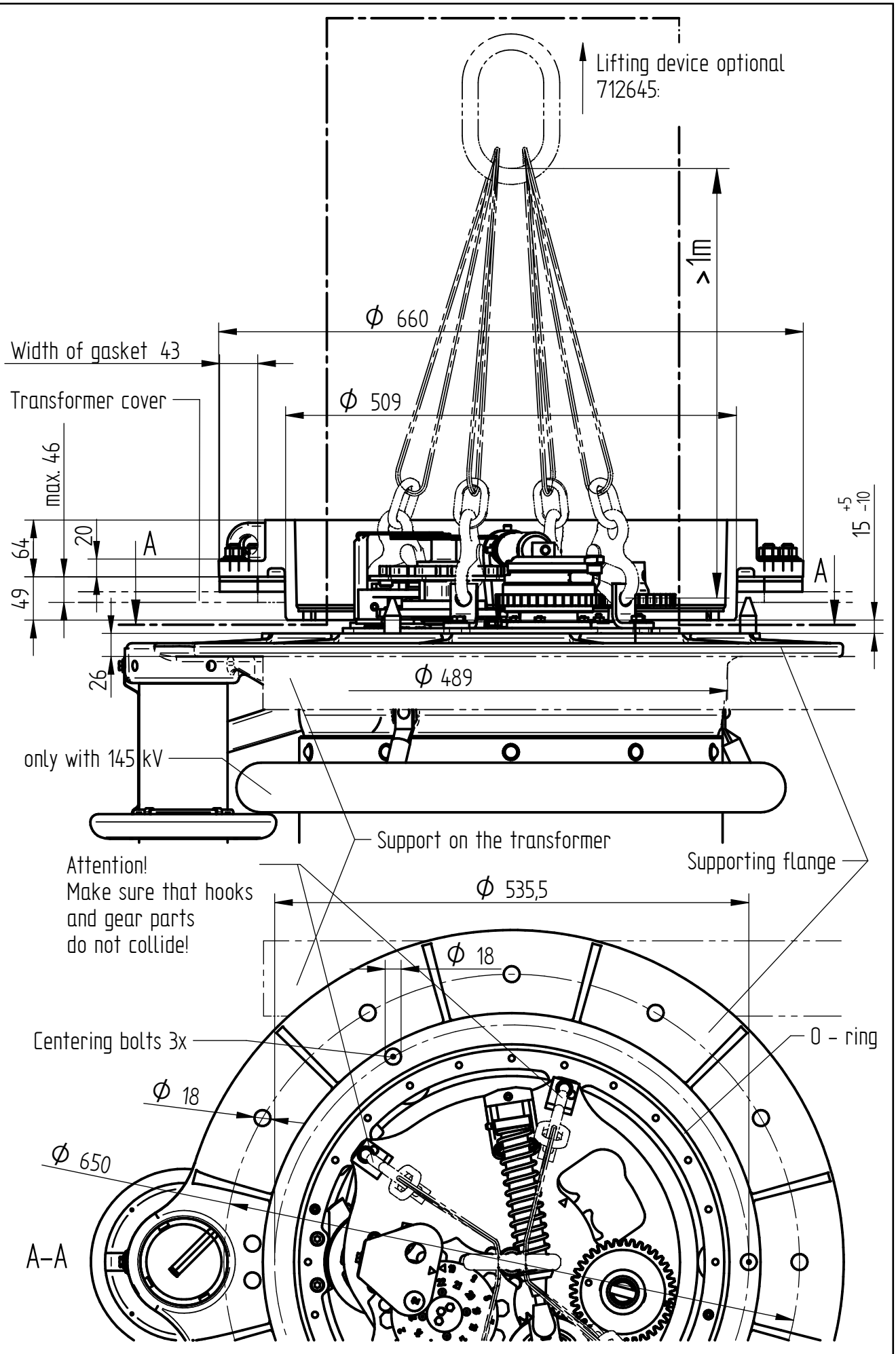
INSTALLATION DRAWING

SERIAL NUMBER –

MATERIAL NUMBER 7389020E

SHEET 1/1

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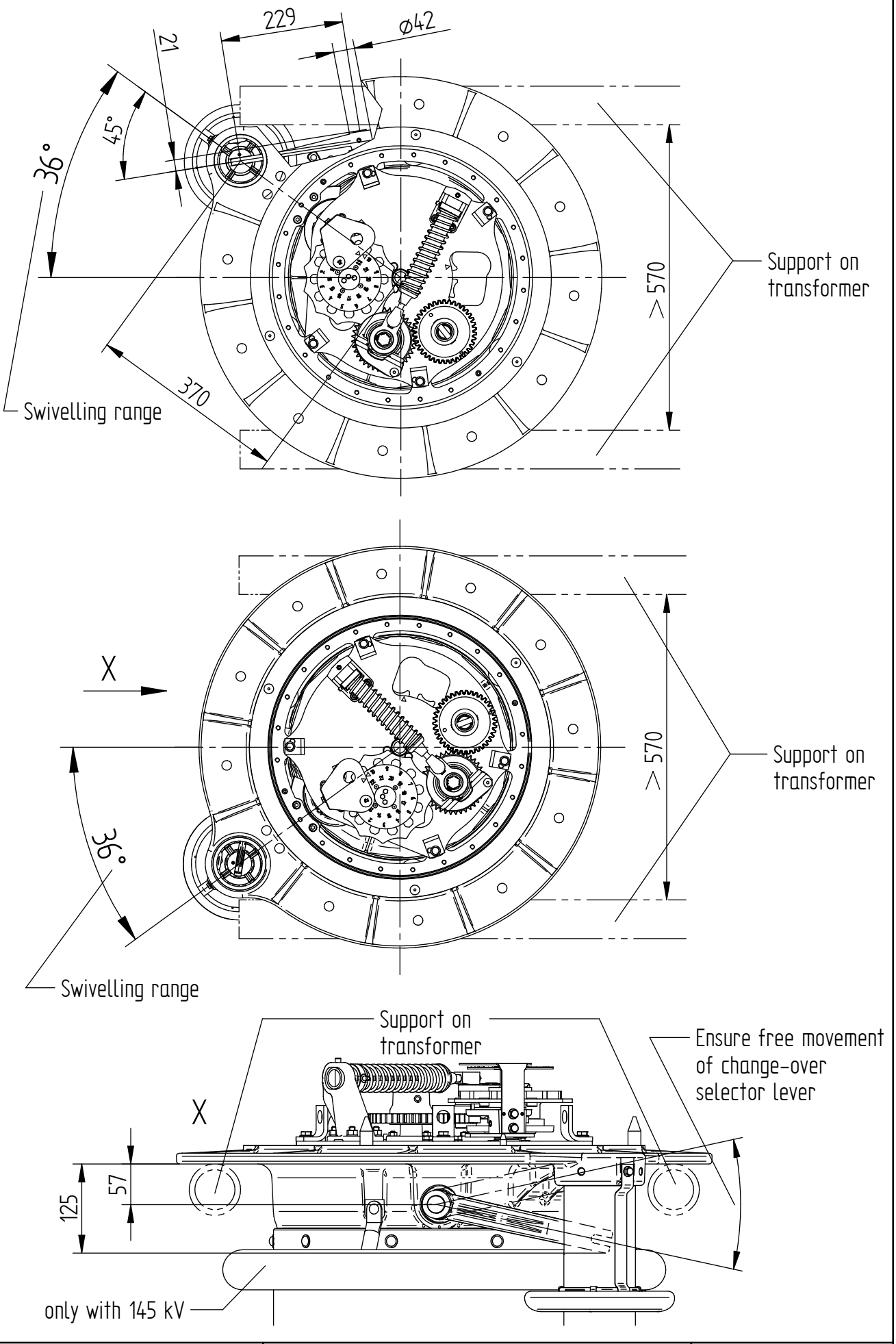
4E	1006204	02.02.06	SPL				
3E	043849	18.12.02	DC				
2E	042782	21.11.02	DC				
1E	035102	05.11.01	EN				
Ae. I	Aend.-Nr:	Tag	Name				
gez. Str 22.11.02							
gepr. Albrecht							

MR

On-Load Tap-Changer VACUTAP® V V

assembly drawing bell-type mounting

8991104E



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3E	1006204	08.02.06	SPL			
2E	043849	18.12.02	DC			
1E	042782	21.11.02	DC			
NEU	037102	05.11.01	EN			
Ae. I	Aend.-Nr.	Tag	Name			
gez.	DC	22.11.02				
gepr.	Albrecht					

MR

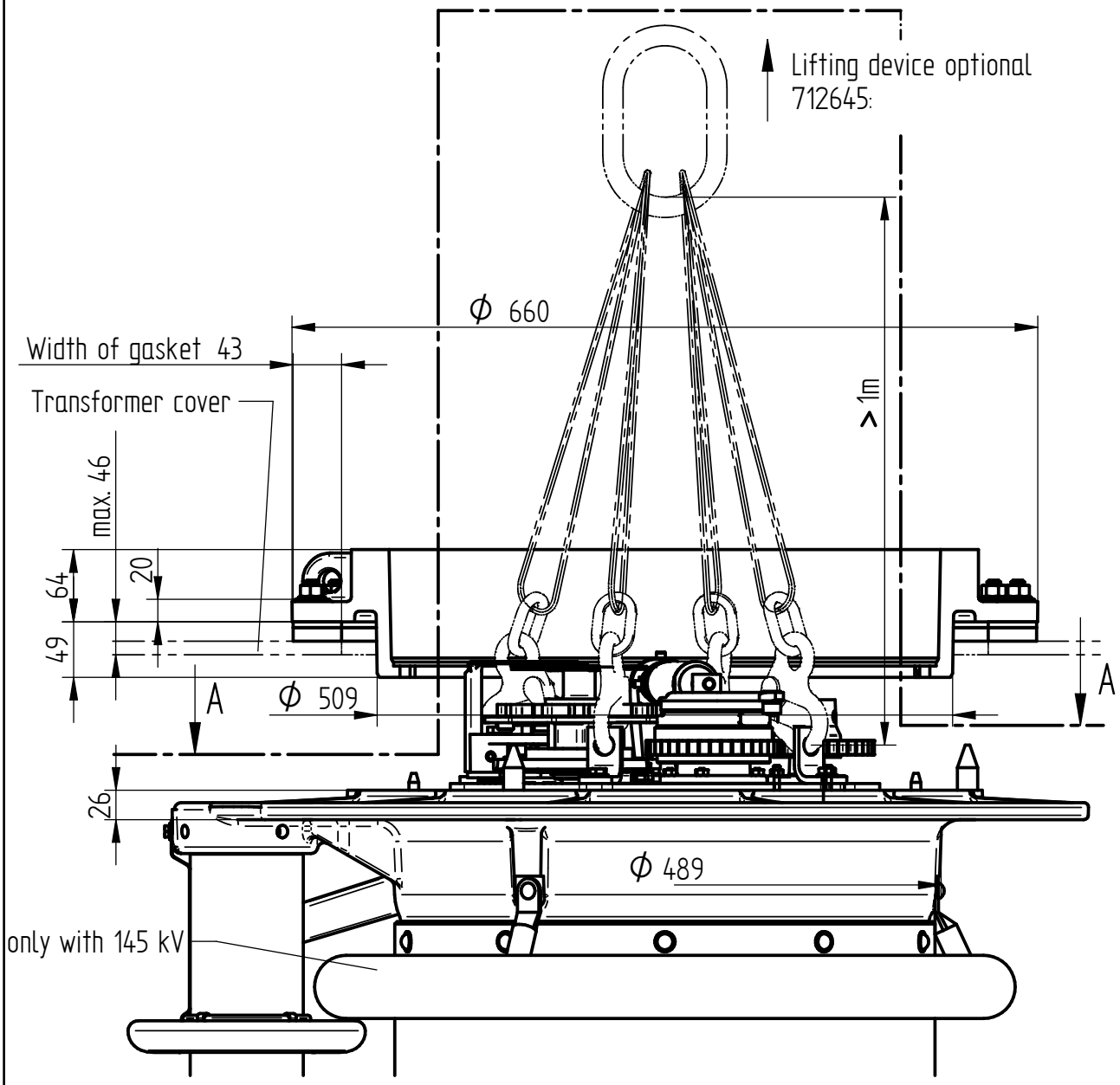
On-Load Tap-Changer VACUTAP® V V

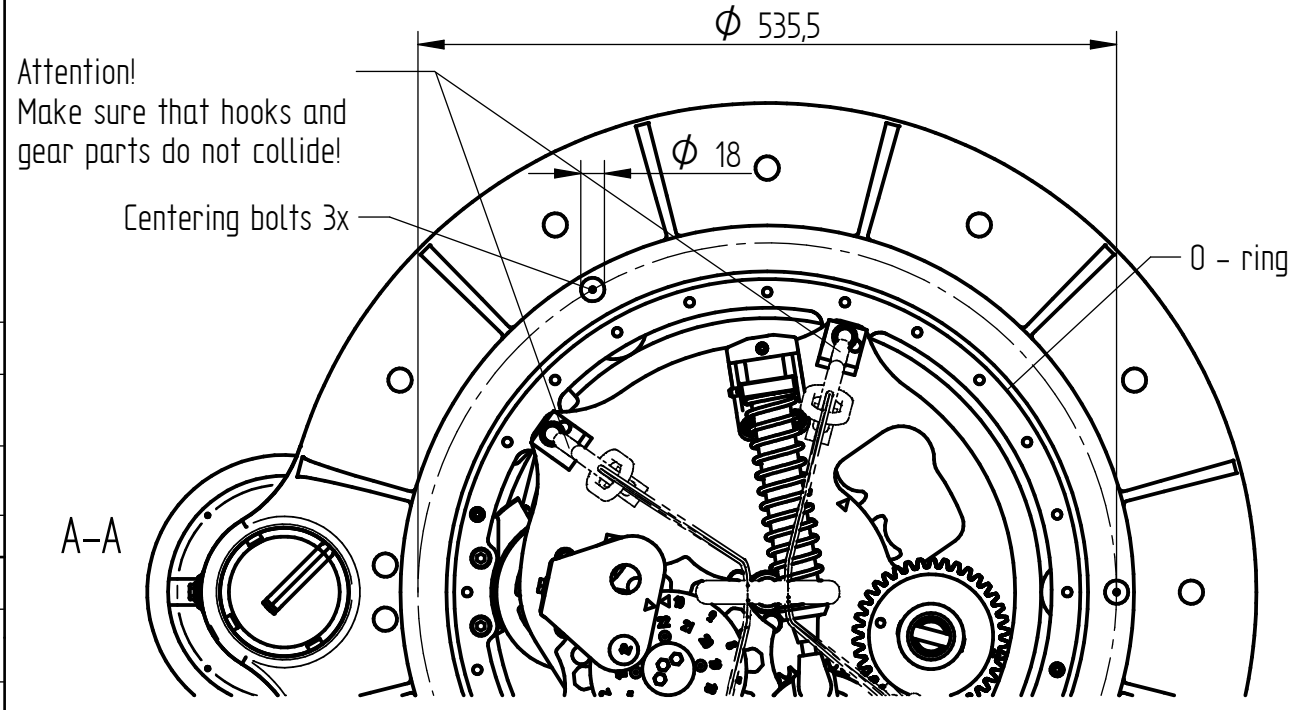
Positioning of OLTC for bell-type mounting

8994093E

DIN A3

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8E	1006204	02.02.06	SPL						
7E	0438449	18.12.02	DC						
6E	0427782	21.11.02	DC						
5E	037102	05.11.01	EN						
Ae.	Aend.-Nr.	Tag	Name	4:0	028788	03.07.00	DC		
gez. Str 22.11.02									
gepr. Albrecht									

MR

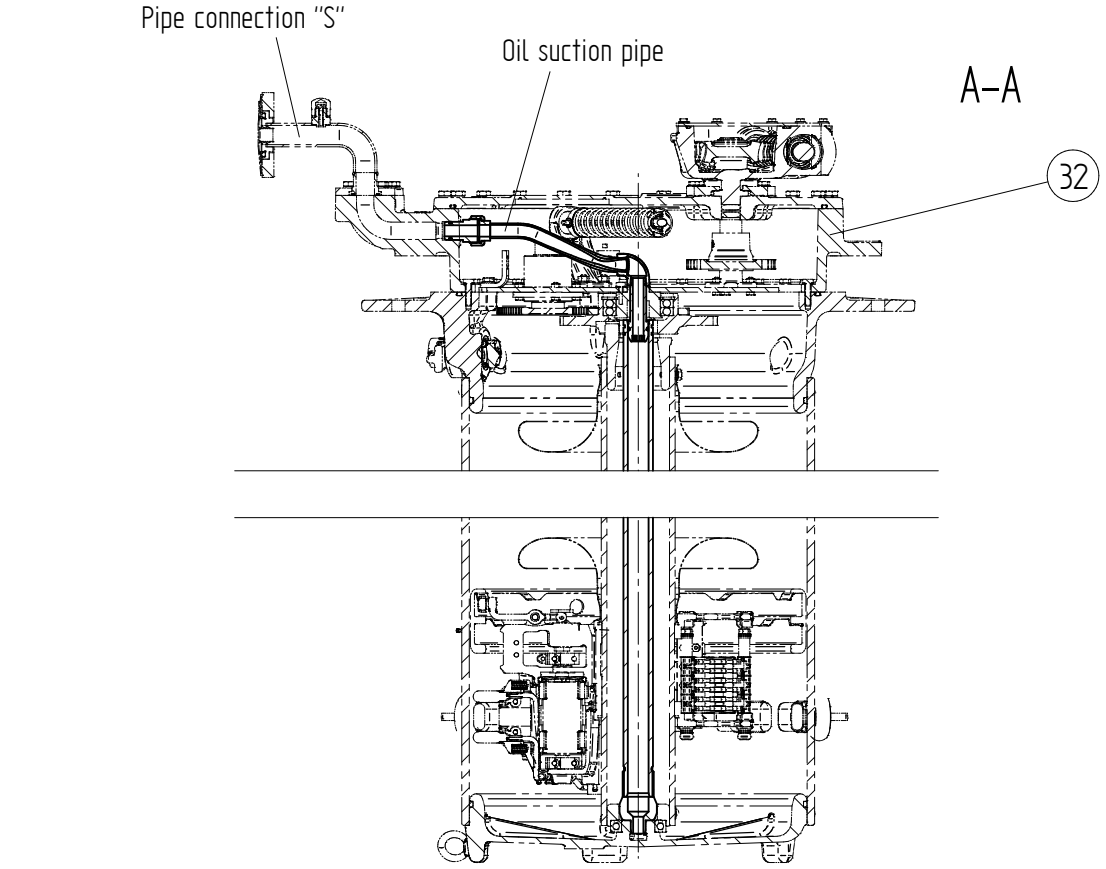
On-Load Tap-Changer VACUTAP® V V

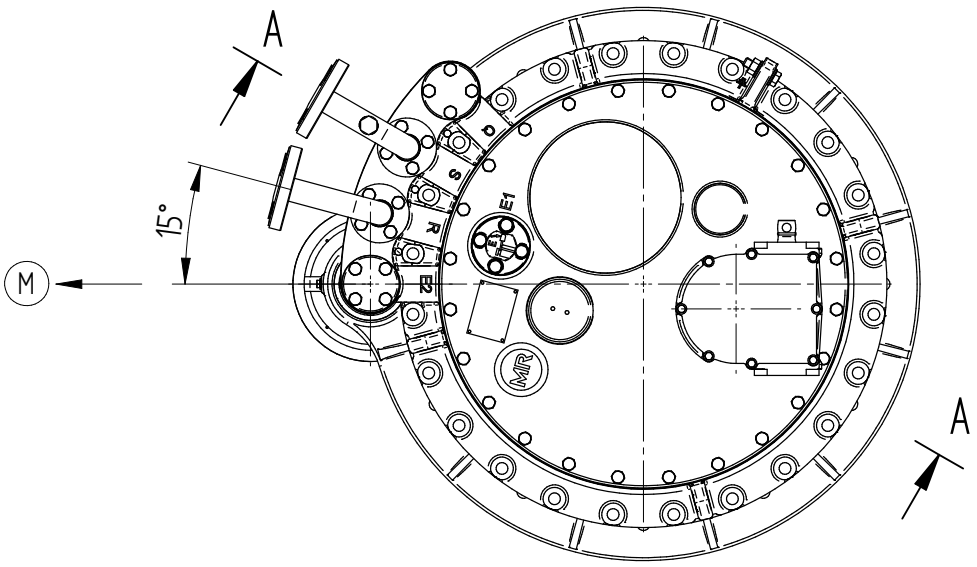
assembly drawing cover mounting

8988668E

3.4 On-load tap-changer head

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(M) Drive side

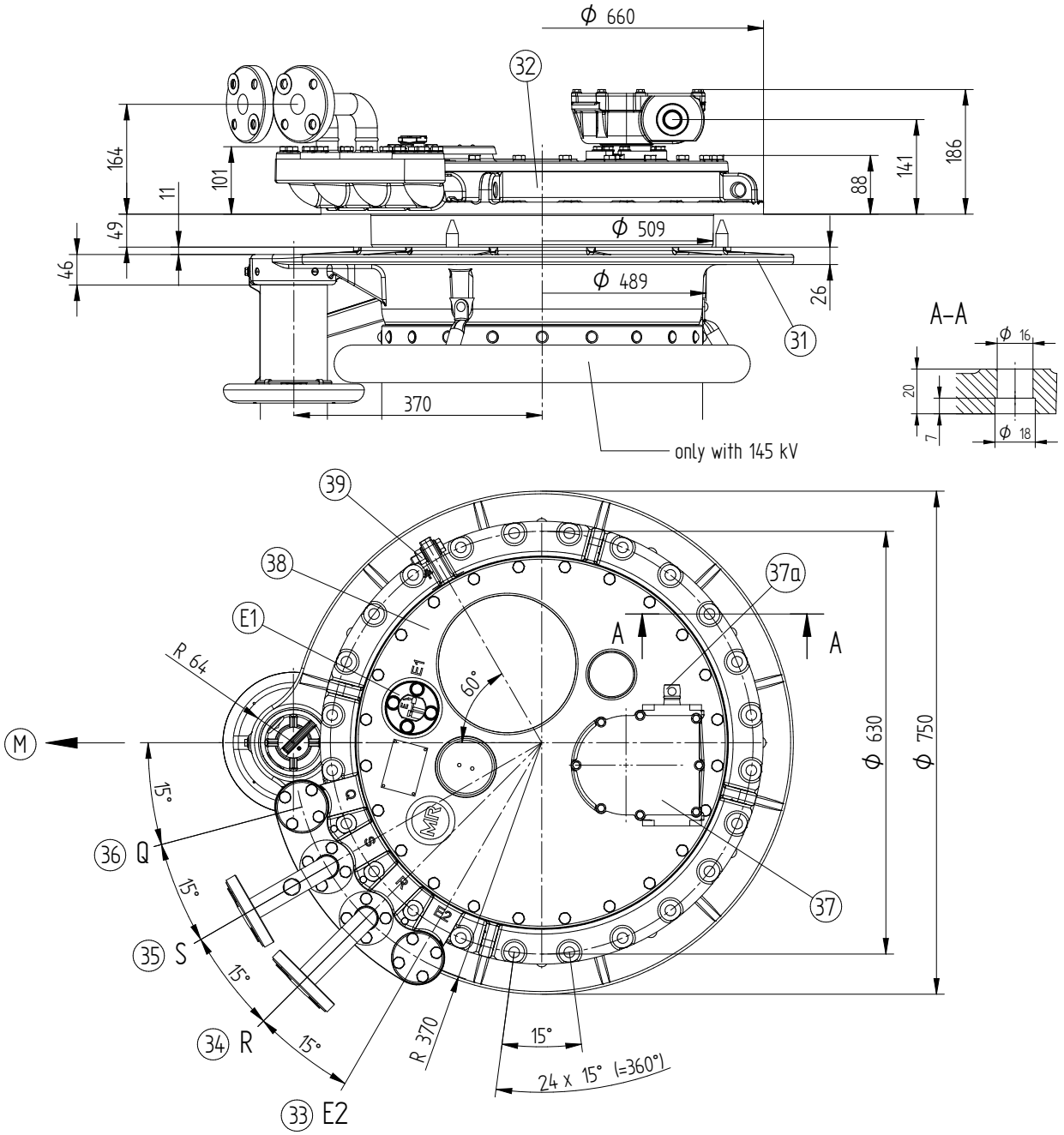
The upper part of the OLTC head (32) CANNOT be turned in the model with oil suction pipe , see drawing 737 060 :

Ch. In	Change No	Date	Name
dftr.	SPL 05.02.09		
chkd			

MR	On-Load Tap-Changer VACUTAP ® V V Oil suction pipe (optional)	739172 0E

DIN A2

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(E1) Bleeding facility for Tap-Changer

(31) Tap-Changer head – bottom part

(32) Tap-Changer head – top part , can be turned towards the bottom part (31) of the head in steps of 15°
The Tap-Changer head cover cannot be turned.

(33) E2 = Bleeding facility for space under the head outside the oil compartment

(34) Pipe connection R for protective relay

(35) Pipe connection S

(36) Pipe connection Q (optional)

Connections can be swivelled 360°
dimensions and selection 899 496 :

(37) Upper gear unit with drive shaft 37a , dimensions and selection see 737 782 :

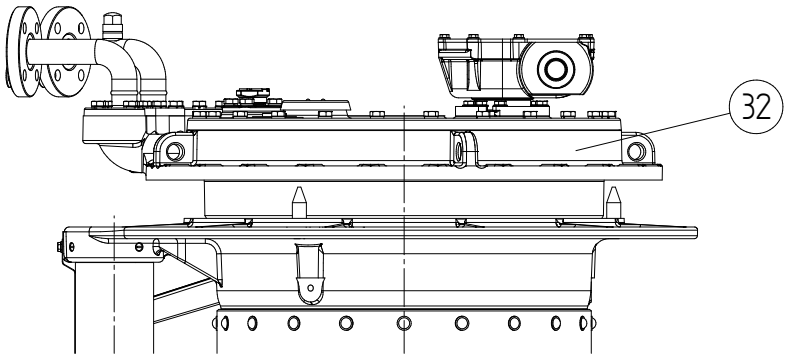
(38) Tap changer head cover

(39) Earth connection M12

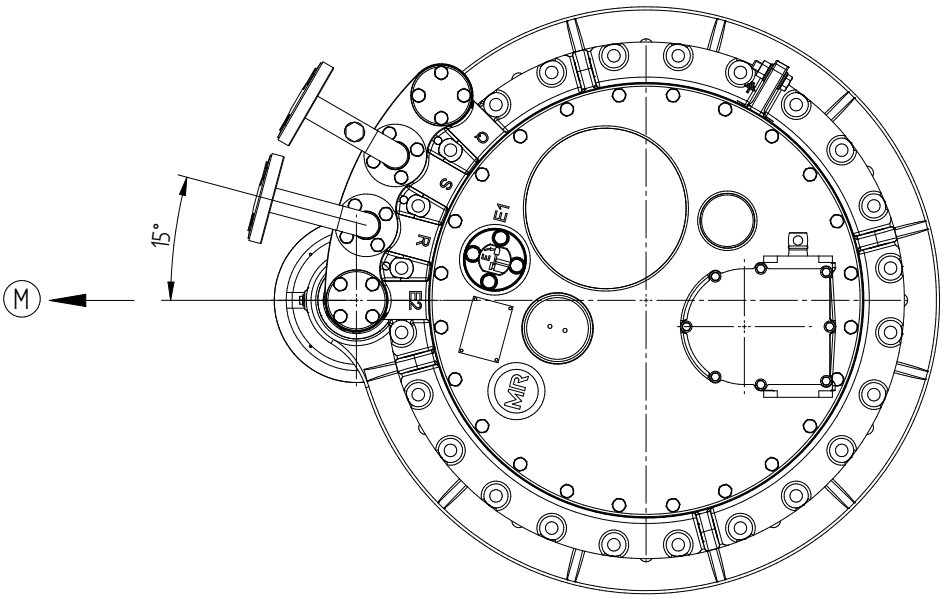
(M) Drive side

⚠ The upper part of the OLTC head (32) CANNOT be turned in the model with oil suction pipe , see drawing 737 060 :

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POSITION OF THE UPPER PART OF THE OLTC HEAD IN THE MODEL WITH OIL SUCTION PIPE

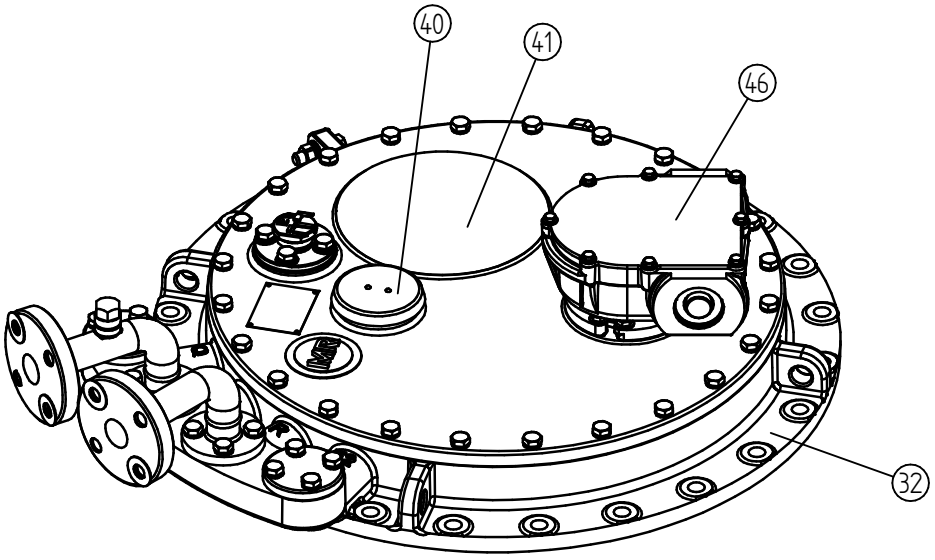


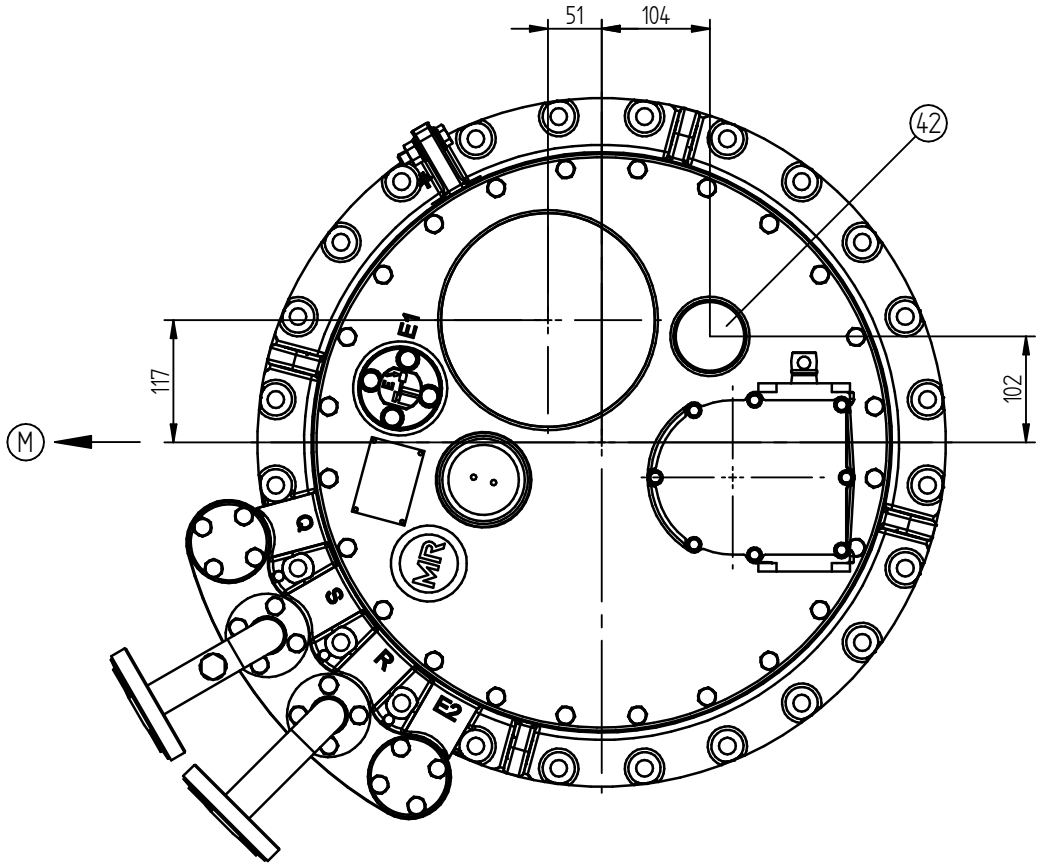
ALL OTHER DIMENSIONS OF THE OLTC HEAD SEE DRAWING 898 863 :

(M) = DRIVE SIDE

⚠ THE UPPER PART OF THE OLTC HEAD (32) CANNOT BE TURNED IN THE MODEL WITH OIL SUCTION PIPE

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(32) Tap-Changer head – top part , can be turned towards the bottom part of the head in steps of 15°
The Tap-Changer head cover cannot be turned.

(40) Inspection glass cover for tap position indicator

(41) Burst pressure membrane

(42) Temperature sensor (optional)

(46) Upper gear unit , dimensions and selection see 737 782 :

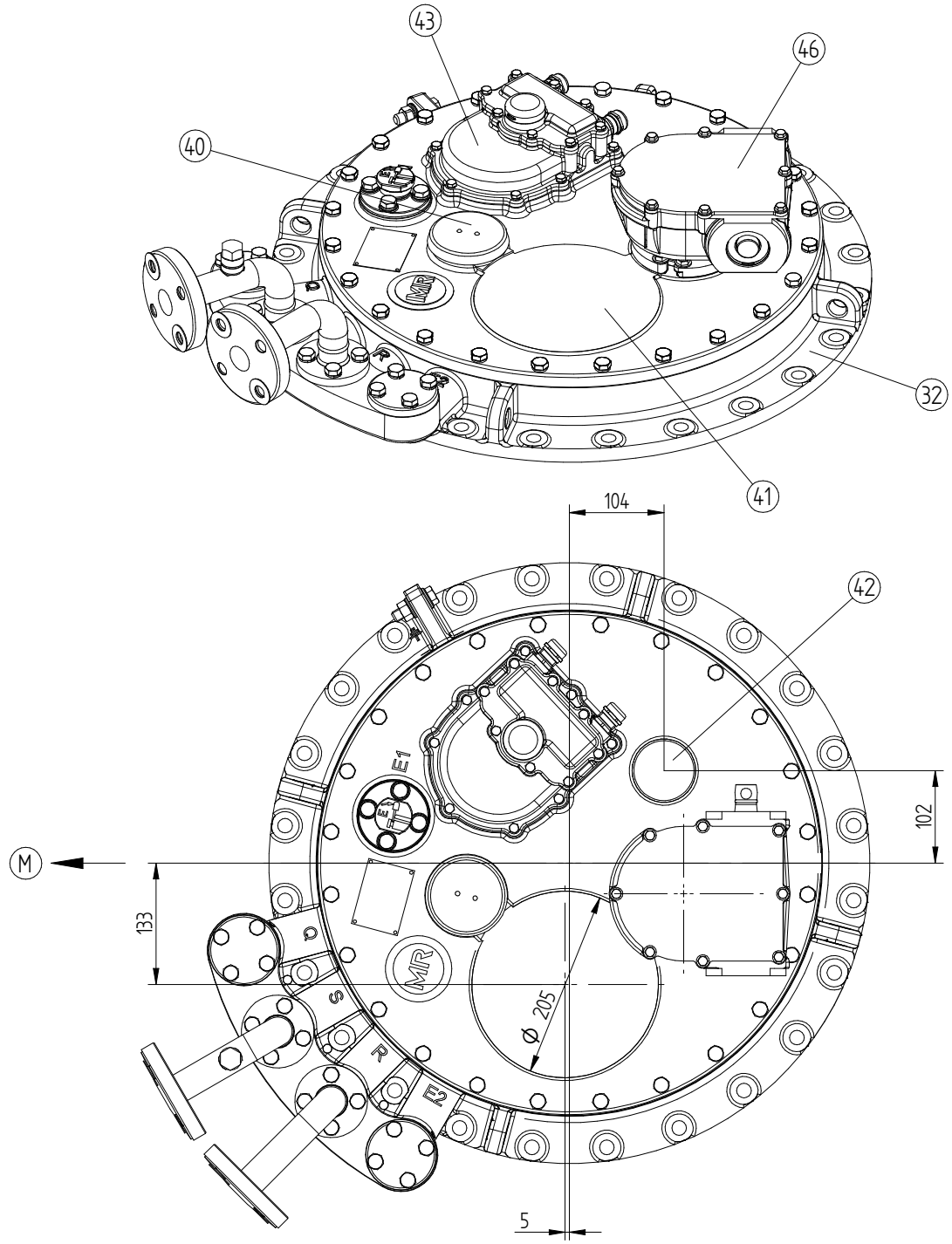
(M) Drive side

The upper part of the OLTC head CANNOT be turned in the model with oil suction pipe , see drawing 737 060 :

	VACUTAP® V V – Head cover design Standard design	737777 0E

DIN A1

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(32) Tap-Changer head – top part , can be turned towards the bottom part of the head in steps of 15°
The Tap-Changer head cover cannot be turned.

(40) Inspection glass cover for tap position indicator

(41) Burst pressure membrane

(42) Temperature sensor (optional)

(43) Supervisory control , dimensions and selection see 733 452 :

(46) Upper gear unit , dimensions and selection see 737 782 :

(M) Drive side

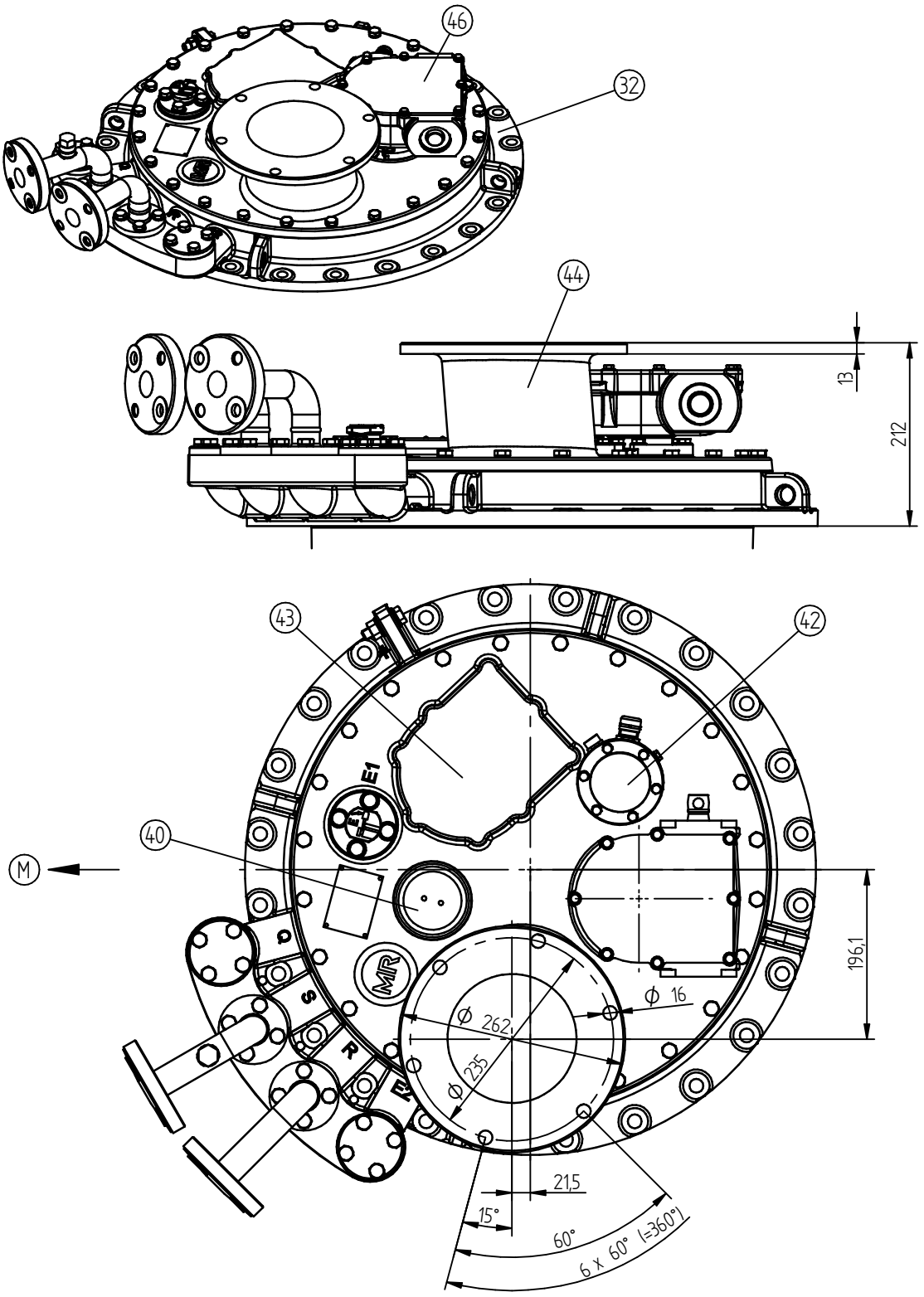
⚠ The upper part of the OLTC head CANNOT be turned in the model with oil suction pipe , see drawing 737 060 :

	VACUTAP® V V – Head cover design Standard design with supervisory control	737778 0E

DIN A1

			02.02.09	SPL
		RYNAS®	Date	Name
		TC		
		JS-19		
diffr. SPL 02.02.09				
chkd				

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(32) Tap-Changer head – top part , can be turned towards the bottom part of the head in steps of 15°
The Tap-Changer head cover cannot be turned.

(40) Inspection glass cover for tap position indicator

(42) Temperature sensor (optional)

(43) Supervisory control , dimensions and selection see 733 452 :

(44) Flange for pressure relief device

(46) Upper gear unit , dimensions and selection see 737 782 :

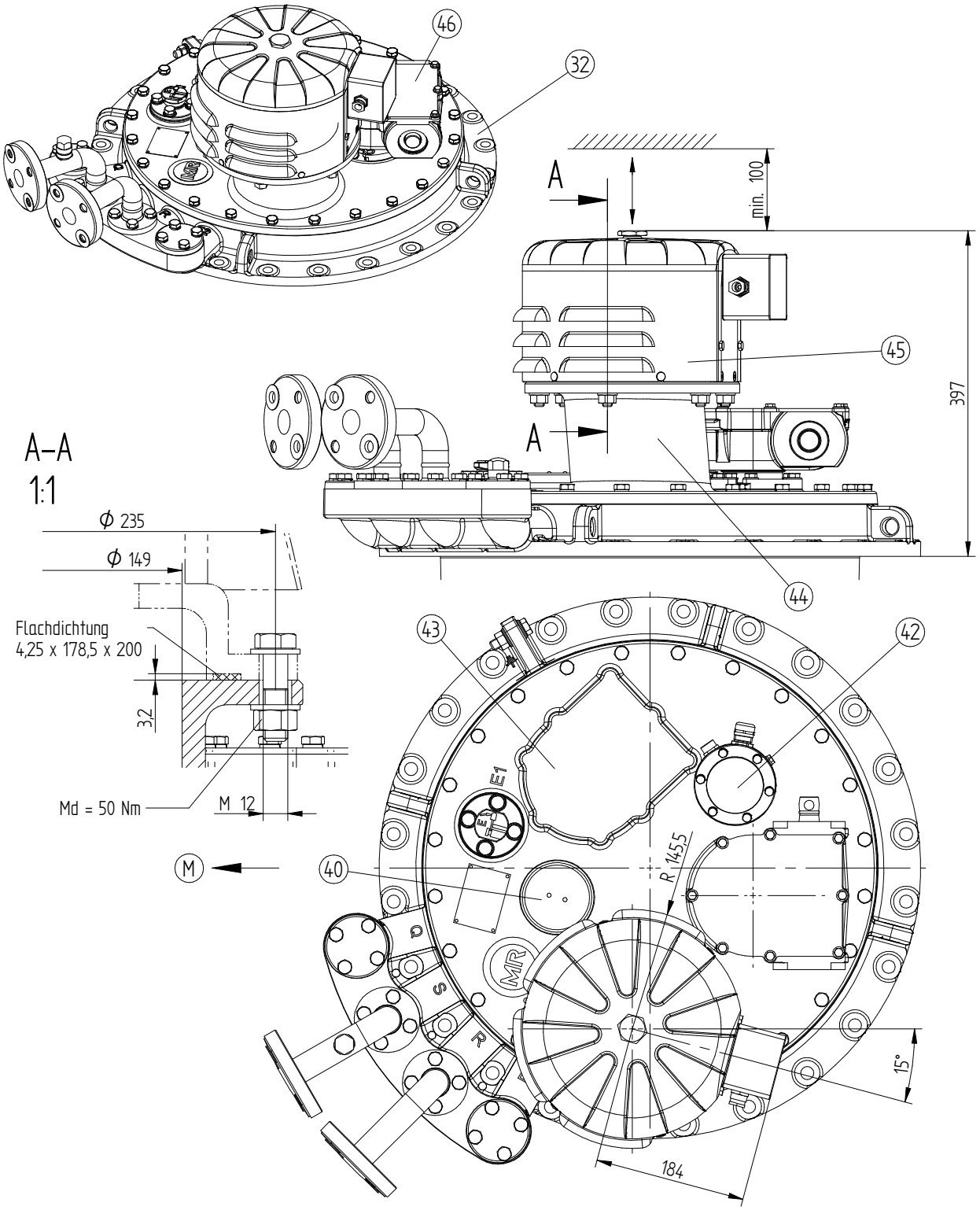
(M) Drive side

The upper part of the OLTC head CANNOT be turned in the model with oil suction pipe , see drawing 737 060 :

	VACUTAP® V V – Head cover design with flange for pressure relief devive	737779 0E

DIN A1

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(32) Tap-Changer head – top part , can be turned towards the bottom part of the head in steps of 15°
The Tap-Changer head cover cannot be turned.

(40) Inspection glass cover for tap position indicator

(42) Temperature sensor (optional)

(43) Supervisory control , dimensions and selection see 733 452 :

(44) Flange for pressure relief device

(45) Pressure relief device , can be installed turned in 60° steps

(46) Upper gear unit , dimensions and selection see 737 782 :

(M) Drive side

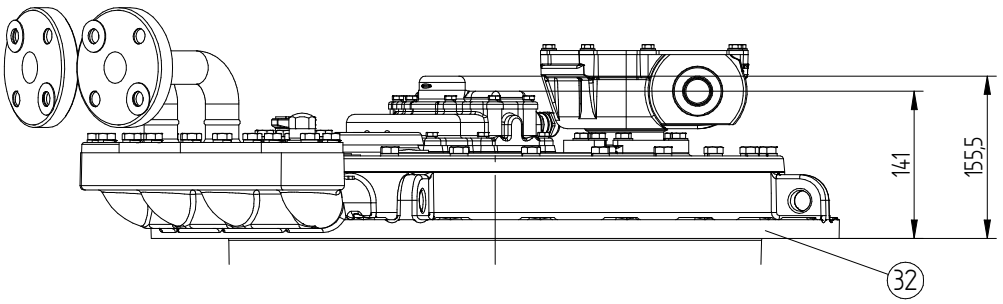
⚠ The upper part of the OLTC head CANNOT be turned in the model with oil suction pipe , see drawing 737 060 :

	VACUTAP® V V – Head cover design with pressure relief device MPREC	737780 0E

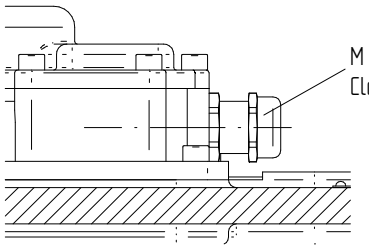
diffr. SPL 04.02.09
chkd

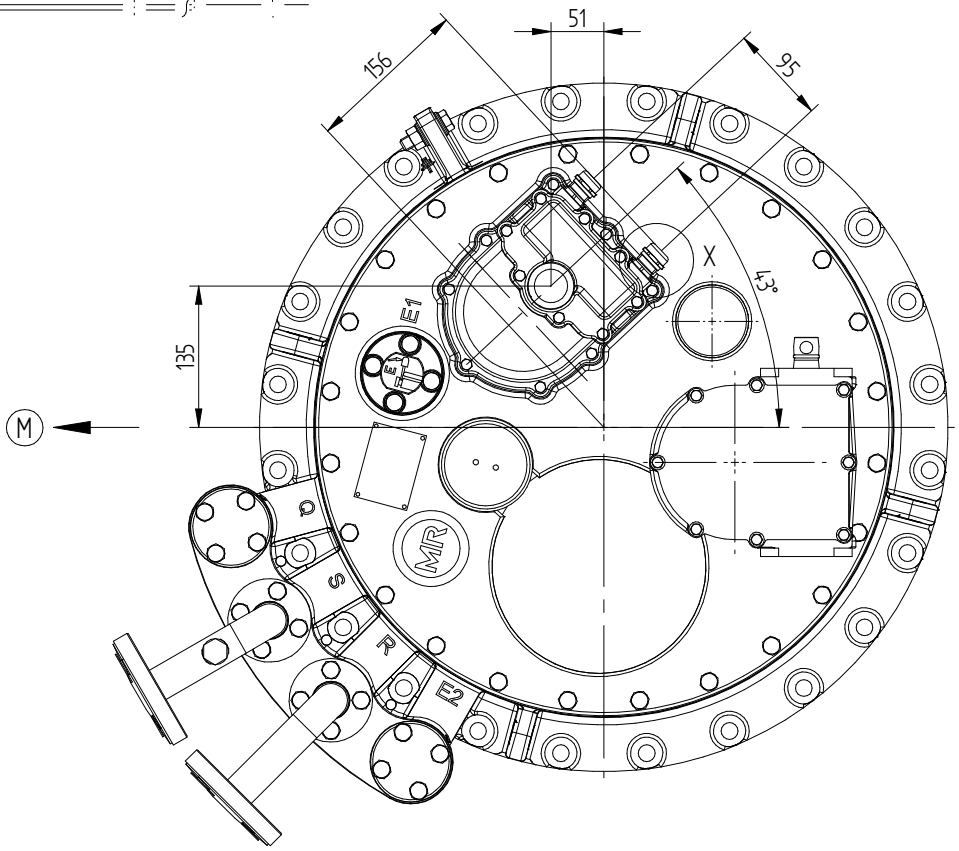
DIN A1

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X
1:1





(32) Tap-Changer head – top part , can be turned towards the bottom part of the head in steps of 15°
The Tap-Changer head cover cannot be turned.

(M) Drive side

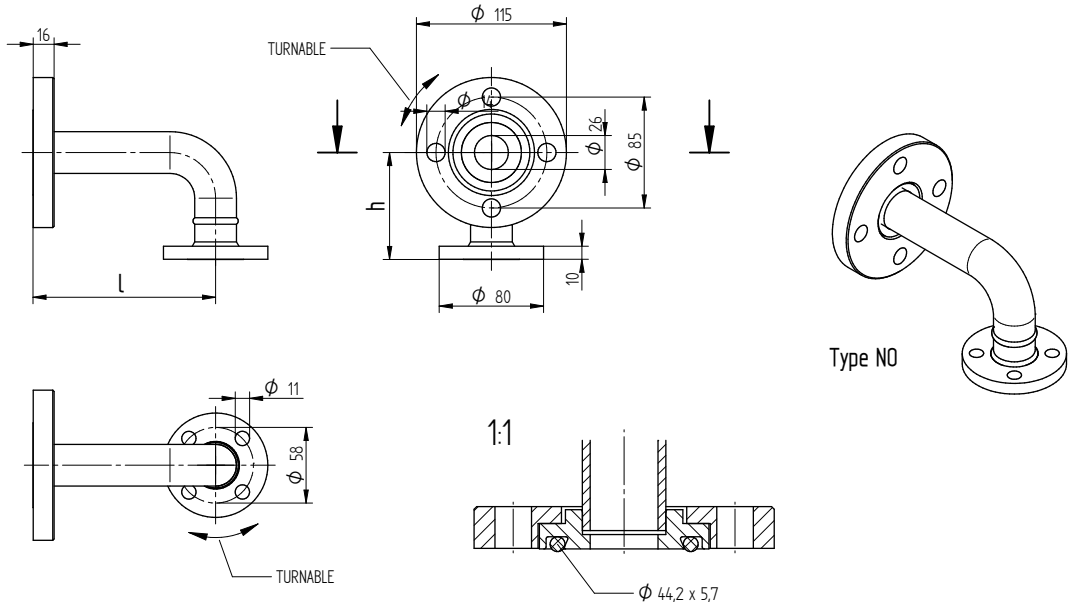
⚠ Take care when installing the electrical lines so that they do not abstruct the OLTC cover when it is opened for maintenance.

⚠ The upper part of the OLTC head CANNOT be turned in the model with oil suction pipe , see drawing 737 060 :

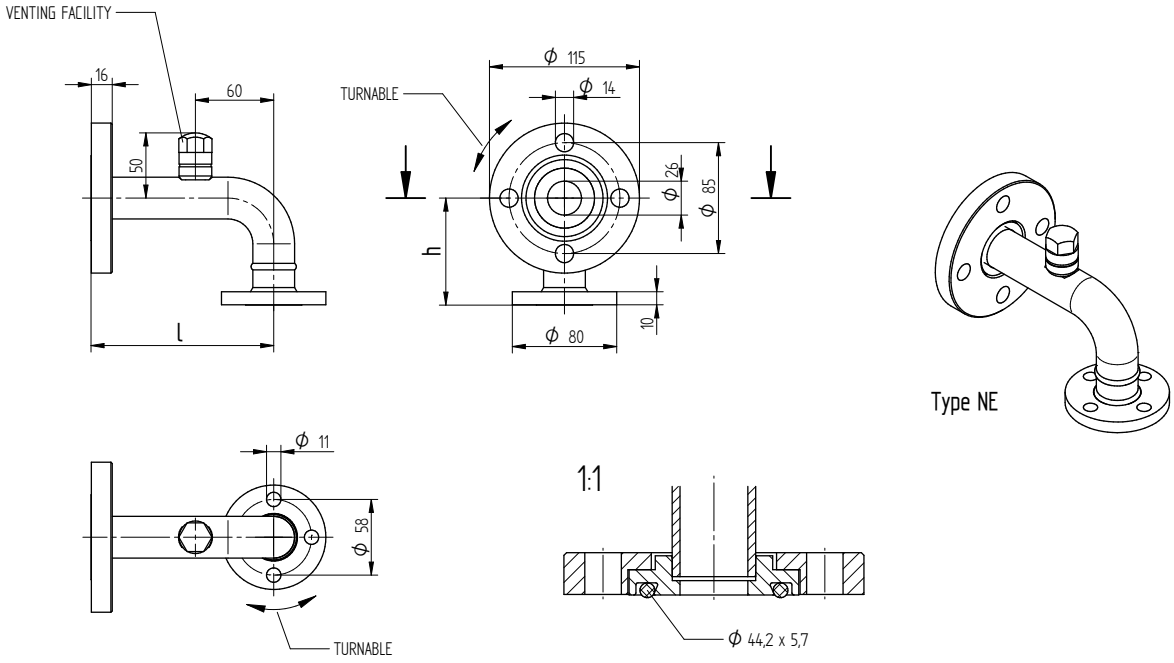
MR	VACUTAP® V V – Head cover design Dimensions of tap change supervisory control	733452 1E

DIN A1

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Type NO



Type NE

EXAMPLE FOR ORDERING A PIPE CONNECTION
WITHOUT VENTING FACILITY WITH l=140 AND h=45 :

" PIPE CONNECTION TYPE NO 140 x 45 "

	STANDARD DIMENSIONS		DESIGN	
	" l "	" h "	TYPE NO	TYPE NE
	140	45	●	
	140	68	●	●
PREFERRED DESIGN	140	82	●	●
	228,6	82		●
	140	132	●	
	140	183	●	
	140	255	●	●
	140	313	●	●

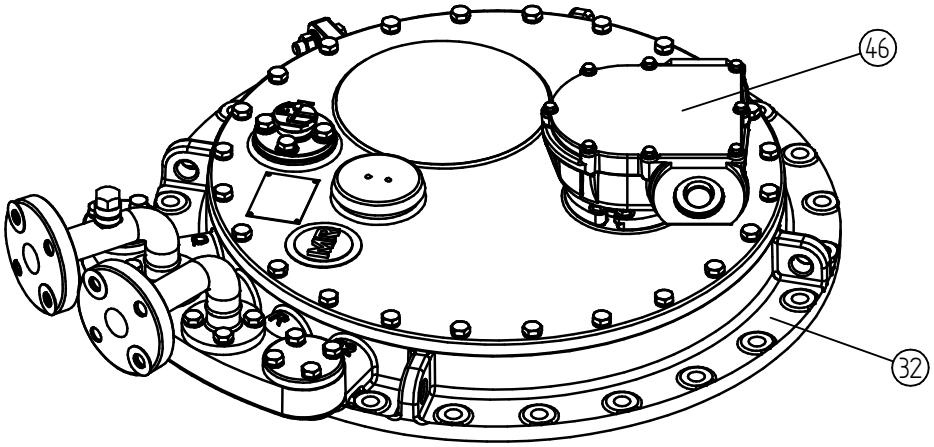
● = AVAILABLE

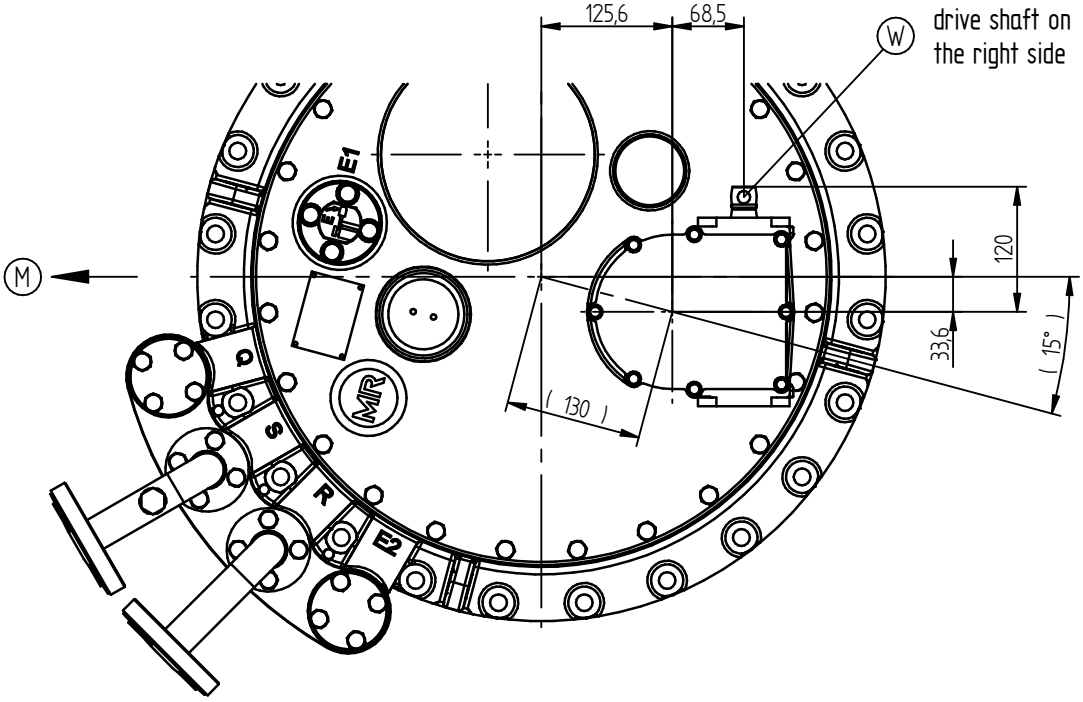
ALL FLANGES ARE TURNABLE BEFORE THE BOLTING

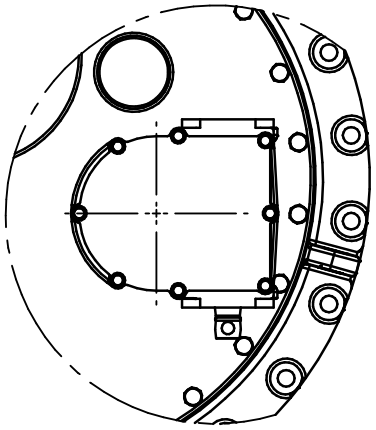
REINHAUSEN GERMANY	PIPE CONNECTION .	8994960E

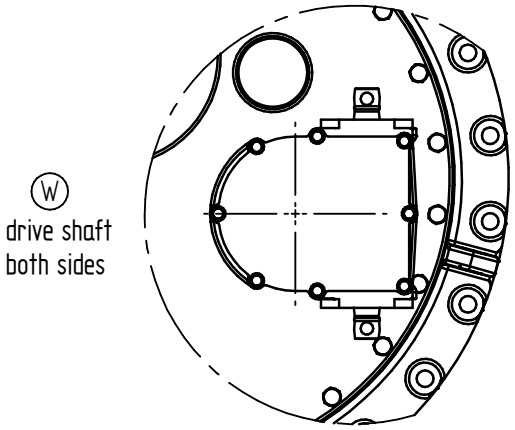
DIN A1

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(32) Tap-Changer head – top part , can be turned towards the bottom part of the head in steps of 15°
The Tap-Changer head cover cannot be turned.

(46) Upper gear unit , steplessy turnable to 360° , Limitation of the swivelling range see 899 210 : and 899 211 :

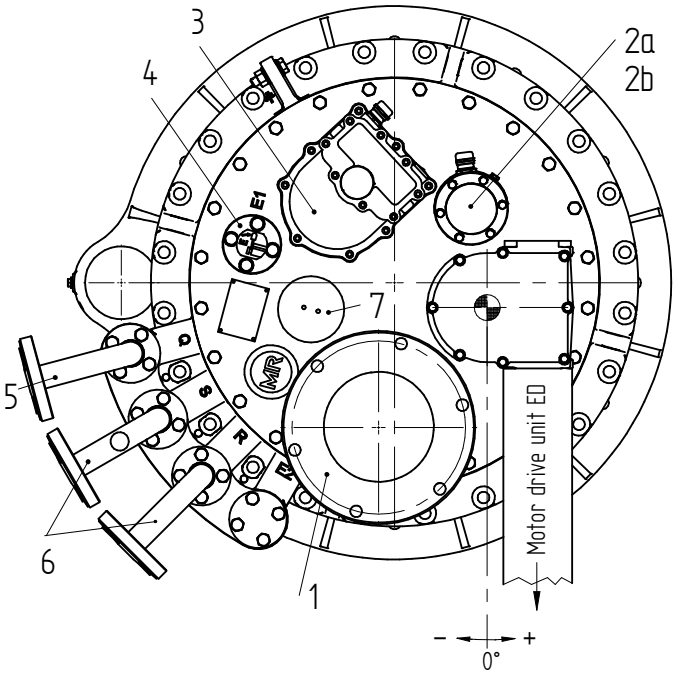
(W) Drive side

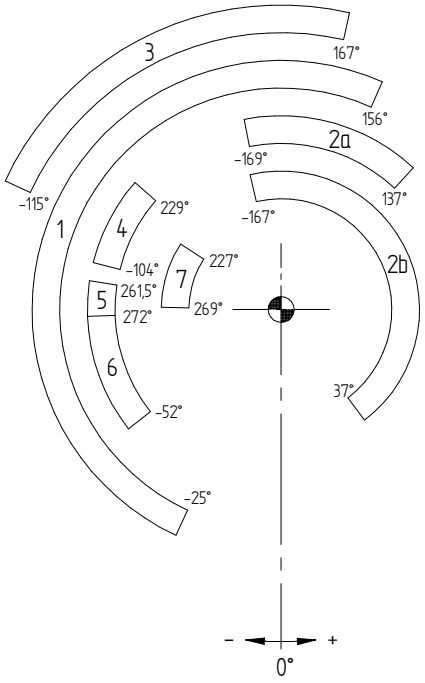
⚠ The upper part of the OLTC head CANNOT be turned in the model with oil suction pipe , see drawing 737 060 :

MR	VACUTAP® V V – Drive shaft Versions of the upper gear unit	737782 0E

DIN A1

Drive shaft on the left :





= Swivel range not possible because of :

1 = Flange for pressure relief device
2a = Temperature sensor or
2b = Thermometer case
3 = Supervisory control
4 = Bleeding facility
5 = Pipe connection "Q"
6 = Pipe connection "S" und "R" (Standard)

7 = This swivel range, which hides the position indication, should be avoided

Swivelling ranges :
By turning around its own axis and by the steplessly turnable upper gear unit, the tap changer can be adapted to the transformer.

The **NOT** usable swivelling ranges of the upper gear unit are shown above on the right.

Only for the design **without** oil suction pipe :
As the top part and the bottom part of the head can be turned to each other (in steps of 15°), the swivelling ranges change accordingly.

The upper part of the OLTC head CANNOT be turned in the model with oil suction pipe , see drawing 737 060 :

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		SPL	Name
	02.04.09	04.02.09	Date
1020699	1016449		Change. No
SE	SE		Ch. In

dftr. SPL 04.02.09
chkd

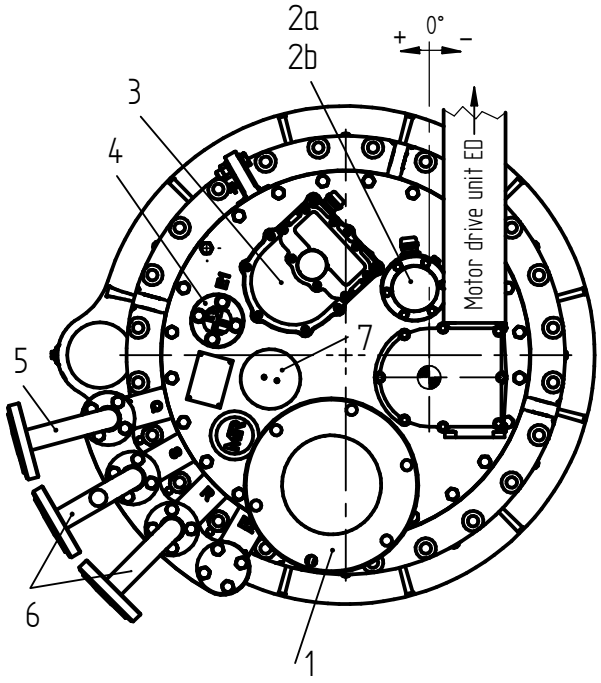
On-Load Tap-Changer VACUTAP® V V

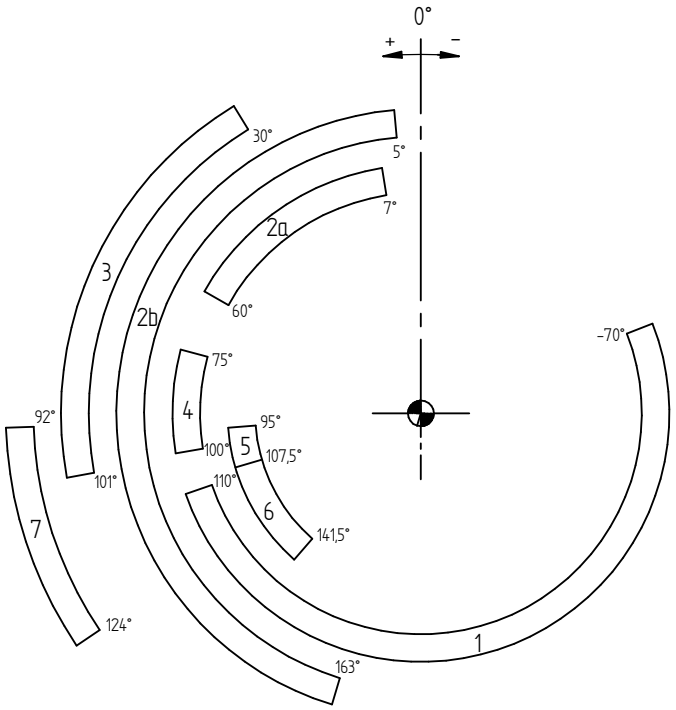
Swivelling range of upper gear unit , drive shaft on the left

899210 5E

DIN A2

Drive shaft on the right :





= Swivel range not possible because of :

1 = Flange for pressure relief device
2a = Temperature sensor or
2b = Thermometer case
3 = Supervisory control
4 = Bleeding facility
5 = Pipe connection "Q"
6 = Pipe connection "S" und "R" (Standard)

7 = This swivel range, which hides the position indication, should be avoided

Swivelling ranges :
By turning around its own axis and by the steplessly turnable upper gear unit, the tap changer can be adapted to the transformer.

The **NOT** usable swivelling ranges of the upper gear unit are shown above on the right.

Only for the design **without** oil suction pipe :
As the top part and the bottom part of the head can be turned to each other (in steps of 15°), the swivelling ranges change accordingly.

The upper part of the OLTC head CANNOT be turned in the model with oil suction pipe , see drawing 737 060 :

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Ch. in	Change No	Date	Name
SE	1020699	02.04.09	SPL
SE	1016449	04.02.09	SPL

dftr. SPL 04.02.09
chkd

MR

On-Load Tap-Changer VACUTAP® V V

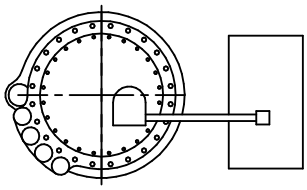
Swivelling range of upper gear unit , drive shaft on the right

899211 5E

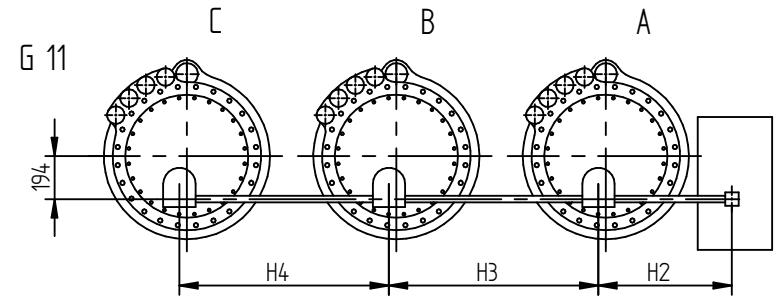
DIN A2

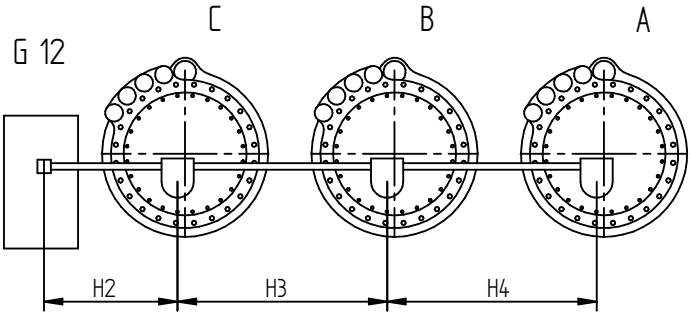
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Arrangement G 4
(Standard design)

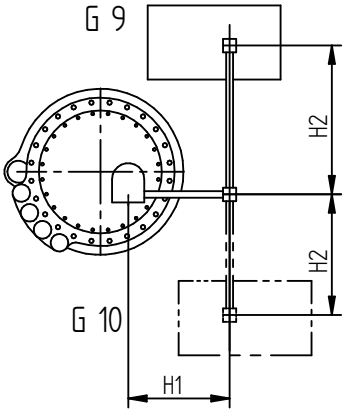


Arrangement G 11 , G 12
(Standard design)

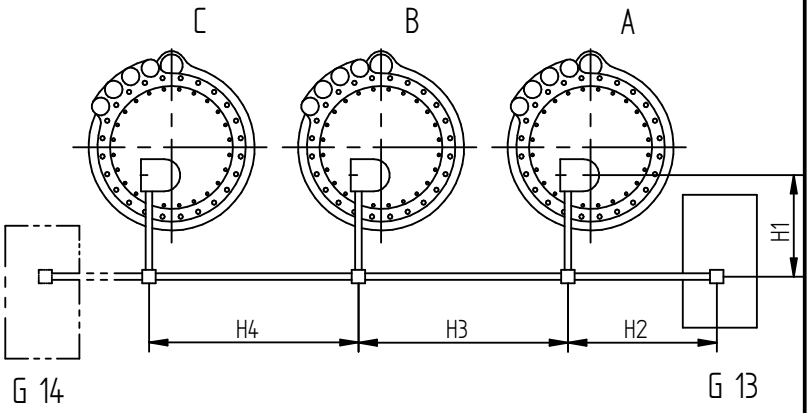




Arrangement G 9 , G 10
(Special design)



Arrangement G 13 , G 14
(Special design)



Minimum dimensions				Intermediate bearing required for		
Arrangement	H1	H2	H3 , H4	H1	H2	H3, H4
G 4	≧ 400	—	Determined by the insulation distance between poles A, B and C (can be mechanically realized up to min. 840 mm)	> 2254	—	—
G 9, G 10	≧ 400	≧ 400		> 2309	> 2259	—
G 11, G 12	—	≧ 400		—	> 2254	> 2249
G 13, G 14	≧ 520	≧ 400		—	> 2259	> 2259

Attention : H1 and H2 can be mechanically realized as minimum dimensions, larger when the upper gear unit is turned into a certain position.

The representations are only examples. It should be checked first if the position (swivel range) of the upper gear unit is possible. See here also 899 210 : und 899 211 :

MR	On-Load Tap-Changerr VACUTAP ® V V Horizontal drive shaft	899169 3E

DIN A2

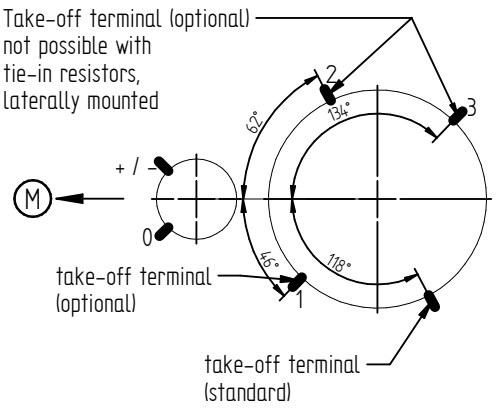
				SPL	Name
			1016449	04.02.09	Date
			3E	Change No	
				Ch. in	
dftr. SPL 04.02.09					
chkd					

3.5 Connection contacts

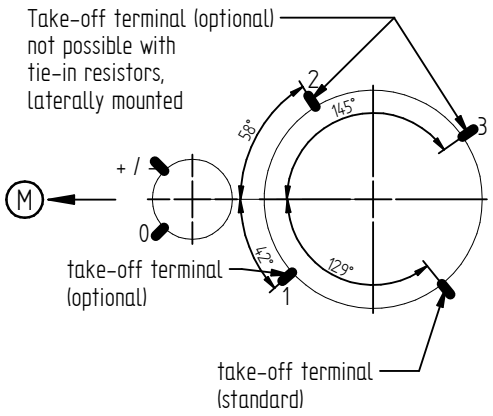
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(21) Location of take-off terminals , Plan view

10 pitch

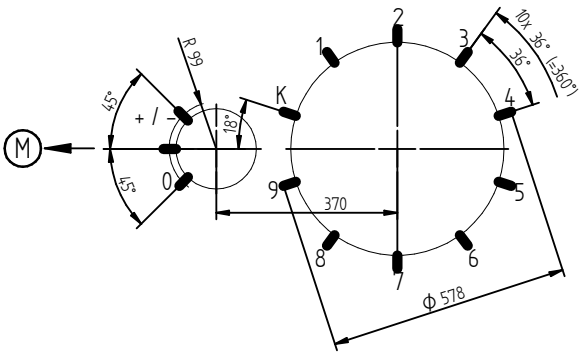


12 pitch

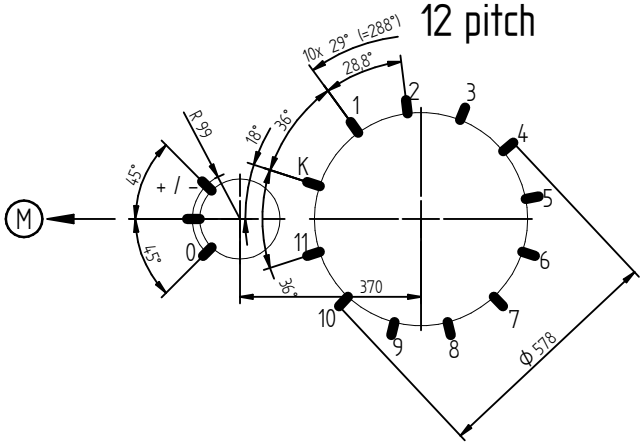


(22) Location of fine tap selector terminals , Plan view

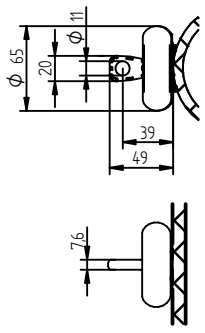
10 pitch



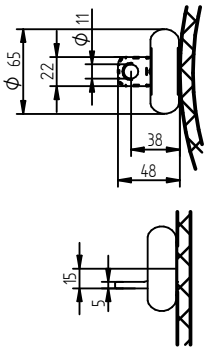
12 pitch



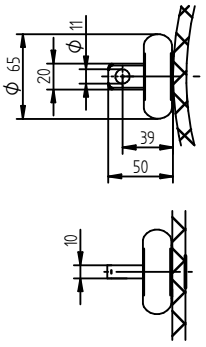
(12) (13) (14) Change-over selector terminal



(21) Take-off terminal



(22) Fine tap selector terminal



(M) = Drive side

The connecting diagram is binding for the designation of the terminals and phases

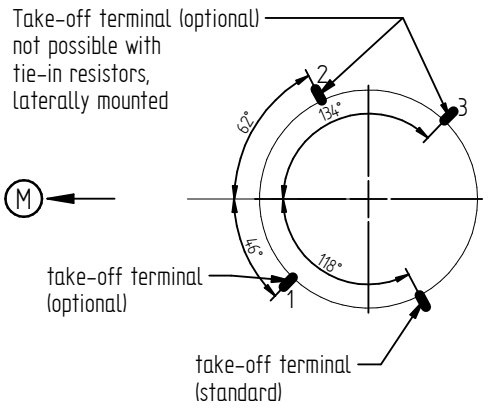
	On-Load Tap-Changer VACUTAP ® V V		
MR	Terminal dimensions - with Change-over selector	737775 0E	

DIN A1

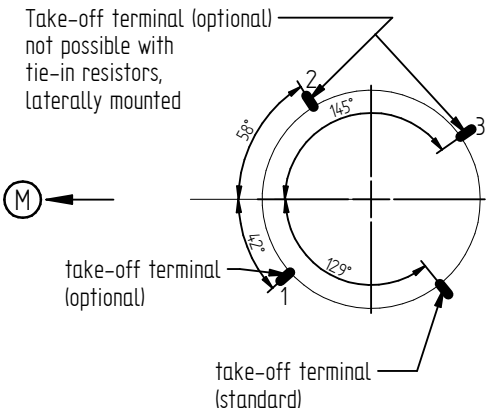
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(21) Location of take-off terminals , Plan view

10 pitch

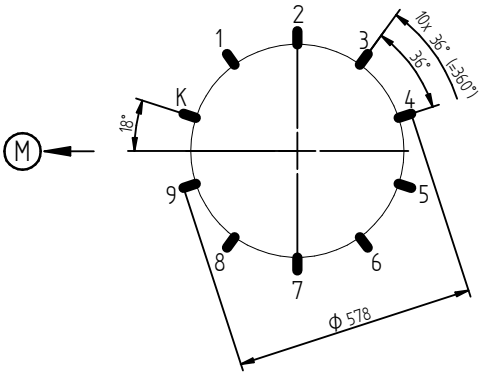


12 pitch

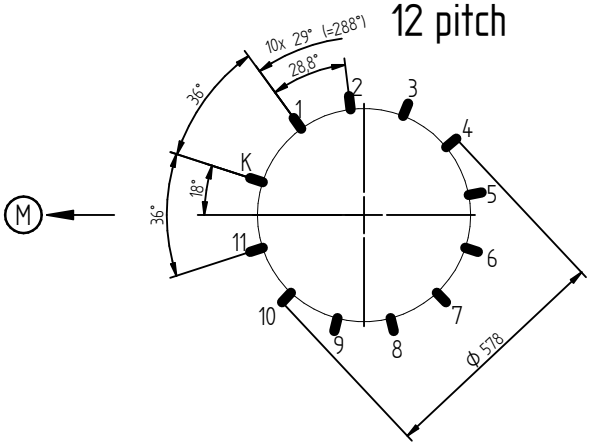


(22) Location of fine tap selector terminals , Plan view

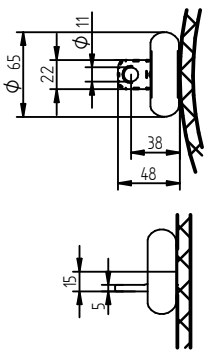
10 pitch



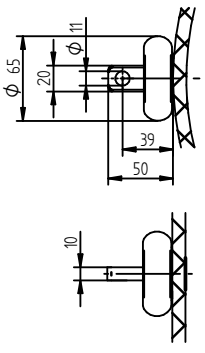
12 pitch



(21) Take-off terminal



(22) Fine tap selector terminal

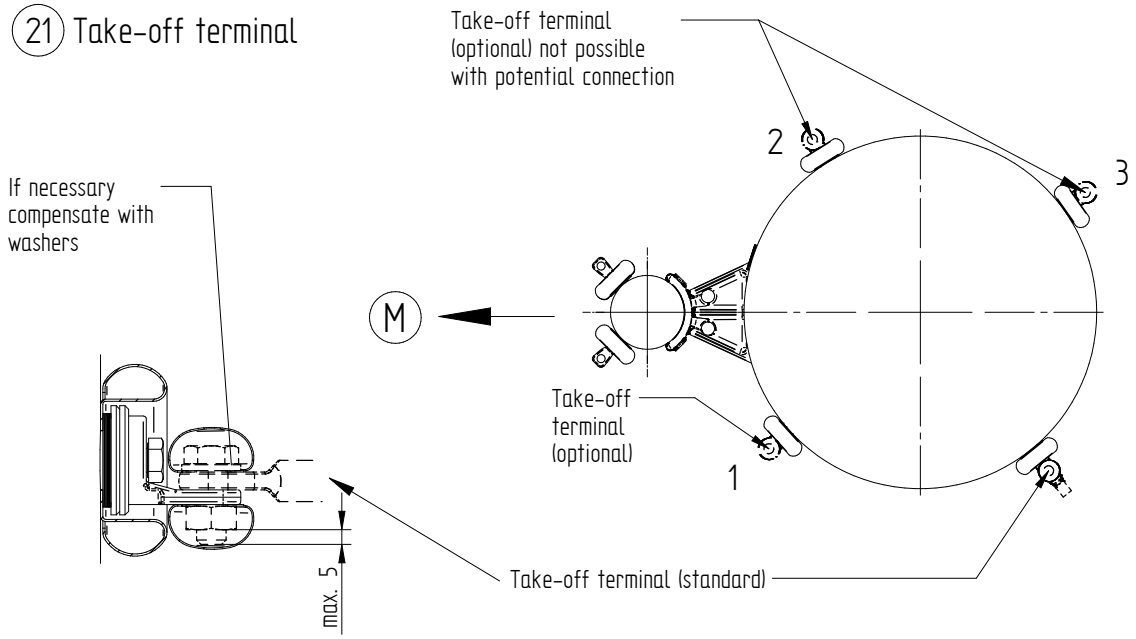


(M) = Drive side

The connecting diagram is binding for the designation of the terminals and phases

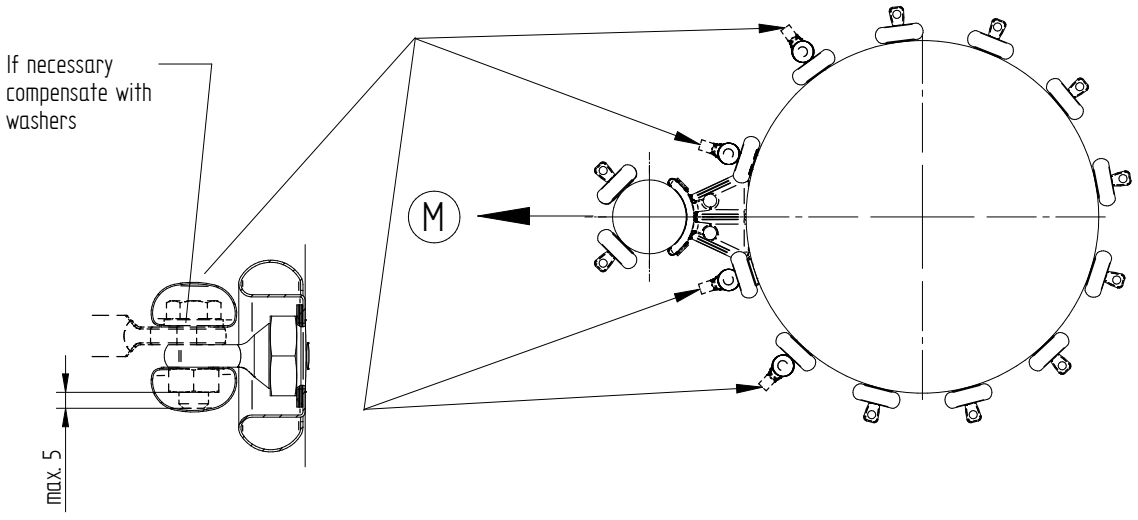
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21 Take-off terminal



These take-off terminals (with or without take-off leads) are to be fitted with screening caps (included in MR's delivery) by the transformer manufacturer.

22 Fine tap selector terminal



These terminals are to be fitted with screening caps (included in MR's delivery) by the transformer manufacturer.

M = Drive side

The connecting diagram is binding for the designation of the terminals and phases

Ch. In	Change No	Date	Name
dftr. Str	210109		
chkd			

MR	On-Load Tap-Changer VACUTAP ® V V Position of the connecting terminals	899051 7E

DIN A2

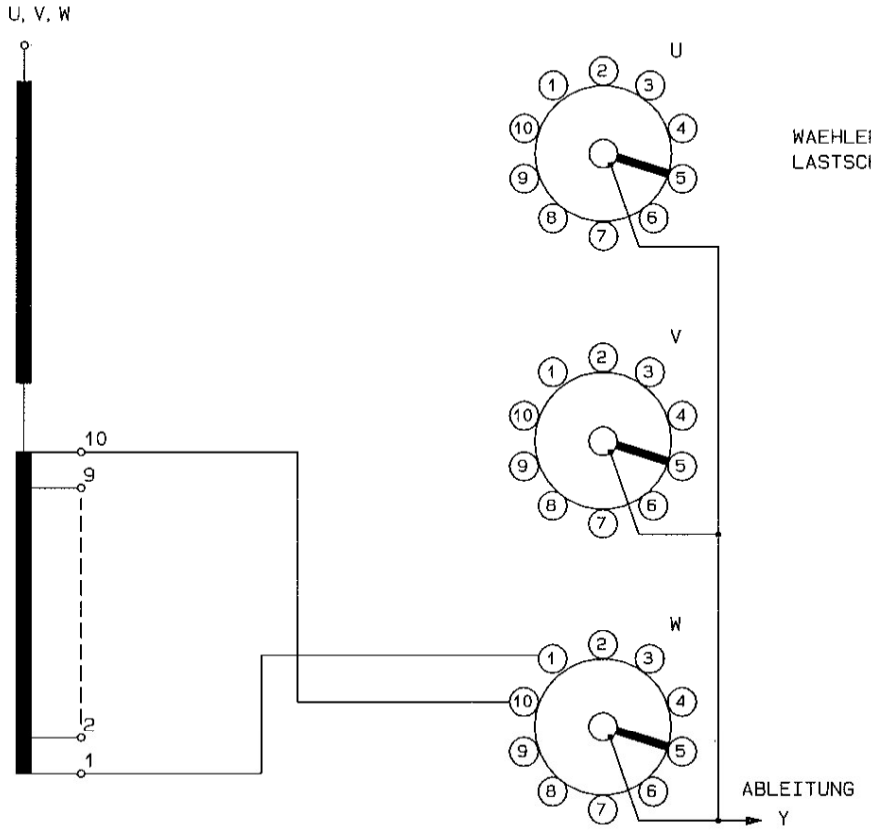
3.6 Connection diagrams (examples)

The connection diagrams contained herein are just examples.

The order-specific connection diagram is enclosed with the delivery.

CAD erstellte Zeichnung
keine manuellen Aenderungen

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WAEHLER/
LASTSCHALTER

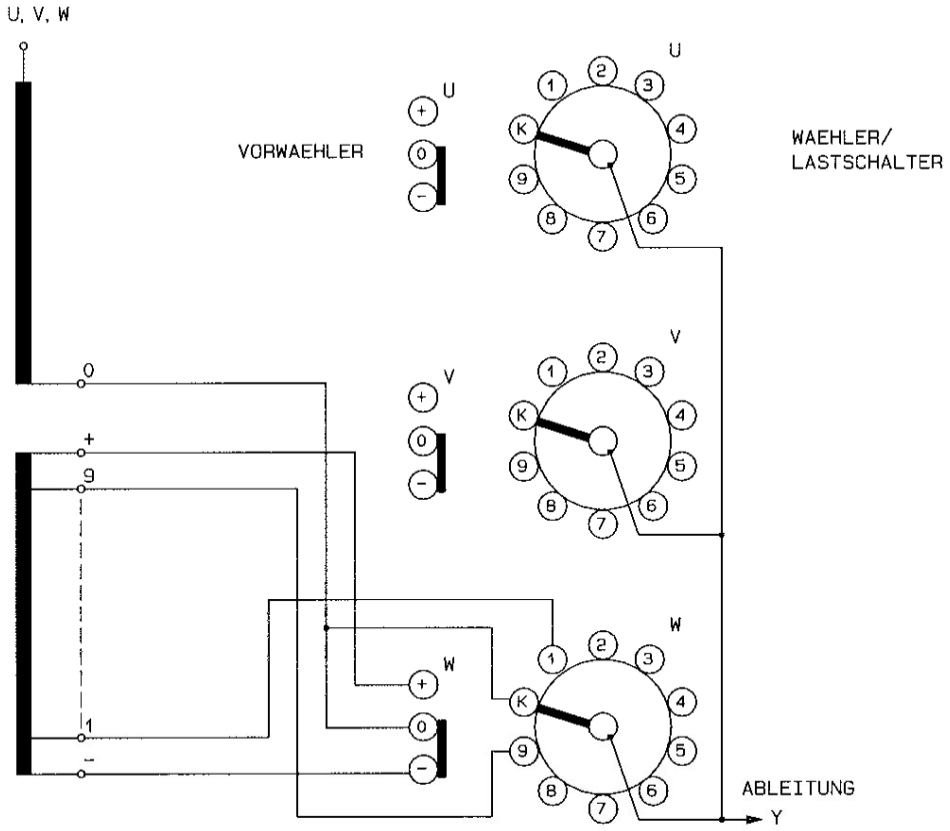
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VERSCHIEDENE SPANNUNGEN	10
JUSTIERSTELLUNG ■	5

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8	8	8
7	7	7
6	6	6
5	5	5
4	4	4
3	3	3
2	2	2
1	1	1

BETRIEBSSTELLUNG
BEZEICHNUNG DER WAEHLERKONTAKTE
BEZEICHNUNG DER STELLUNGEN

| | | |

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BETRIEBSSTELLUNGEN	19
VERSCHIEDENE SPANNUNGEN	19
JUSTIERSTELLUNG ■	10

STELLUNG DES WENDERS		BETRIEBSSTELLUNG		BEZEICHNUNG DER WAEHLERKONTAKTE		BEZEICHNUNG DER STELLUNGEN
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						18
						17
						16
						15
-	0					14
						13
						12
						11
						10
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						8
						7
+	0					6
						5
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						3
						2
						1

				Datum	22.11.02				
				Bearb.	AU				
				Gepr.	BLE				
Zust.	Aenderung	Datum	Name	Norm		Ursprung	Ers.f.	Ers.d.	

MR

LASTSTUFENSCHALTER

VV III Y-10191W

DE

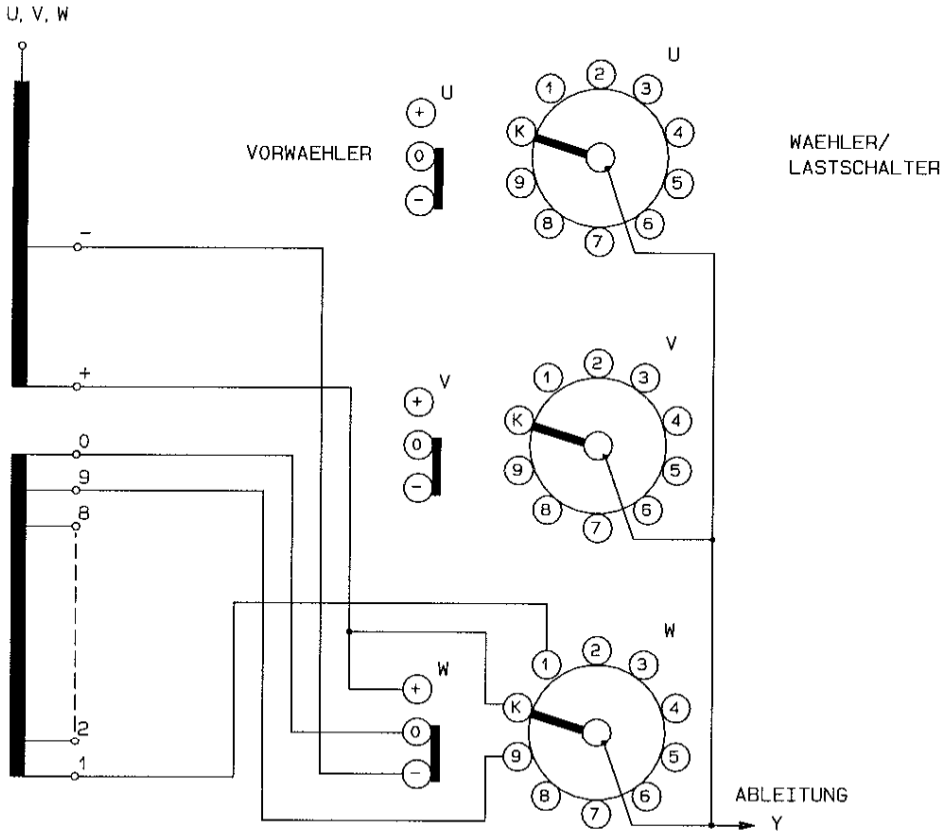
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CAD erstellte Zeichnung
keine manuellen Aenderungen



BETRIEBSSTELLUNGEN	19
VERSCHIEDENE SPANNUNGEN	19
JUSTIERSTELLUNG ■	10

STELLUNG DES GROBWAEHLERS		
BETRIEBSSTELLUNGEN		
BEZEICHNUNG DER WAEHLERKONTAKTE		
BEZEICHNUNG DER STELLUNGEN		
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- 0 +

Zust.	Aenderung	Datum	Name	Norm		Ursprung		Ers.f.		Ers.d.
				Datum	22.11.02					
				Bearb.	AU					
				Gepr.	BLE					

LASTSTUFENSCHALTER

VV III Y-10191G

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THE POWER BEHIND POWER.